EVIDENCE REVIEW • ASIA AND THE PACIFIC

Welfare Losses from Aggregate Socioeconomic and Environmental Shocks

A multidimensional review of COVID-19, economic crises, disasters, environmental degradation, and climate change

Prepared as a near-publication-ready chapter for internal review

Economic Research and Development Impact Department

Evidence cutoff: 31 July 2026

Review corpus: 52 quantitative studies and assessments

Scope note

Observed losses, causal estimates, accounting assessments, exposure measures, and long-run scenarios are labelled separately. Estimates are not summed across incompatible units or overlapping populations.

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Abstract

Major aggregate shocks in Asia and the Pacific have generated welfare losses that are larger, more unequal, and more persistent than headline changes in gross domestic product suggest. This review synthesizes quantitative evidence on the COVID-19 pandemic, economic shocks since 2015, and environmental and climate shocks, interpreting welfare through the capability approach, human-development framework, and welfare economics. The evidence indicates that COVID-19 was the largest synchronized regional shock in the review period: developing Asia lost an estimated 6.0-9.5% of regional GDP relative to a no-pandemic baseline in 2020; 75-80 million additional people were pushed into extreme poverty; and Asia-Pacific employment declined by an estimated 81 million jobs (Asian Development Bank, 2021a, 2021b; International Labour Organization, 2020). Its durable burden is concentrated in excess mortality, untreated illness, learning loss, nutrition, mental health, and foregone earnings. Economic shocks were more heterogeneous but locally severe. The Sri Lankan crisis approximately doubled poverty between 2021 and 2022, while the Lao inflation and currency crisis produced large real-wage and food-security losses (World Bank, 2023b, 2024a). Across low- and middle-income countries, a 5% increase in real food prices raised the risk of child wasting by 9%, illustrating how cost-of-living shocks are converted into human-capital damage (Headey and Ruel, 2023). Environmental shocks are recurrent rather than episodic: floods, cyclones, heat, drought, fire, and pollution destroy assets and also reduce consumption, productivity, health, learning, and resilience. In small Pacific economies, single events produced damage and losses equal to 31% of GDP in Fiji and 64% in Vanuatu. Long-run climate scenarios imply much larger cumulative risks, but are less certain than observed disaster and microeconometric estimates. Across shock types, children face the longest duration of loss; working-age adults carry employment, earnings, migration, and care burdens; and older people experience the highest mortality and care-related losses. Evidence is strongest for mortality, employment, poverty, learning, disaster damage, food-price nutrition effects, and heat-productivity effects, but remains weak on multidimensional welfare aggregation, older-person income security, disability, gendered time use, informal coping, and intergenerational transmission. The central policy implication is that resilience should be evaluated by preservation of capabilities and consumption, not merely by restoration of output.

Keywords: welfare loss; COVID-19; inflation; food prices; climate change; disasters; human capital; poverty; Asia and the Pacific

Reading the estimates

Ranges preserve the original scenario or uncertainty interval. GDP gaps, asset losses, poverty changes, deaths, exposure counts, and lifetime earnings are not summed. Confidence applies to the estimate's use in this review.

1. Introduction

Asia and the Pacific experienced a dense sequence of aggregate shocks after 2015: earthquakes and tropical cyclones, the COVID-19 pandemic, school and health-service closures, supply bottlenecks, commodity-price spikes, monetary tightening, debt distress, conflict, extreme heat, drought, floods, wildfire smoke, and accelerating climate change. These shocks differed in origin and duration, but shared a common feature. Each damaged not only current production but also the substantive freedoms that enable people to live healthy, secure, educated, and socially connected lives. A development assessment restricted to GDP therefore misses much of the welfare burden and can mis-rank both shocks and policy priorities.

This review adopts a multidimensional definition of welfare rooted in the capability approach and the human-development tradition (Sen, 1999; United Nations Development Programme, 1990). Income and consumption remain essential because they support command over goods and services, but they are treated as means as well as ends. Mortality, morbidity, mental health, nutrition, learning, employment quality, security, social inclusion, agency, and resilience are intrinsically valuable capabilities. Welfare economics contributes the counterfactual discipline needed to distinguish loss from a low level: the relevant quantity is the outcome under the shock compared with the outcome that would plausibly have prevailed without it. This distinction is crucial in a rapidly growing region. A positive observed growth rate can coexist with a large counterfactual loss; conversely, direct asset damage can exceed annual GDP without implying an equivalent annual fall in consumption.

The review asks eight related questions. Which shocks generated the largest short-run and long-run welfare losses? Which welfare dimensions and population groups were most affected? Which countries and subregions experienced the most severe setbacks? Through which mechanisms did losses persist? How comparable are the metrics used in the literature? How credible are the underlying identification strategies? What evidence gaps remain? And what policy architecture follows when resilience is understood as the capacity to protect capabilities before, during, and after a shock?

The answer is not a single league table. COVID-19 dominates the recent record as a synchronized regional shock, while disasters dominate the upper tail of country-level losses, particularly in Pacific small island developing states. Inflation and macroeconomic crises often cause smaller regional output losses but impose intense and regressive real-consumption losses on households near subsistence. Environmental degradation generates large annual health losses without a single event date. Climate change raises the frequency and intensity of these burdens and creates potentially enormous long-run losses, but estimates become more assumption-dependent with the horizon. A valid comparison must therefore preserve distinctions among realized and projected losses, stocks and flows, income and capability measures, and directly observed and modelled outcomes.

1.1 Conceptual pathways

Figure 1 organizes the evidence around four stages. First, the initiating shock may be epidemiological, economic, environmental, or compound. Second, it propagates through labour markets, prices, public finance, service delivery, assets, health exposure, ecosystems, and mobility. Third, households and firms respond through savings, borrowing, asset sales, reduced food intake, withdrawal from school or care, additional work, migration, and reliance on social networks. Fourth, those responses alter economic welfare, health, human capital, social inclusion, agency, and resilience. Feedback loops matter. Asset depletion weakens the capacity to absorb the next shock; malnutrition impairs learning and productivity; school withdrawal lowers lifetime earnings; illness reduces labour supply; and fiscal stress can convert a temporary macro shock into persistent service deterioration.

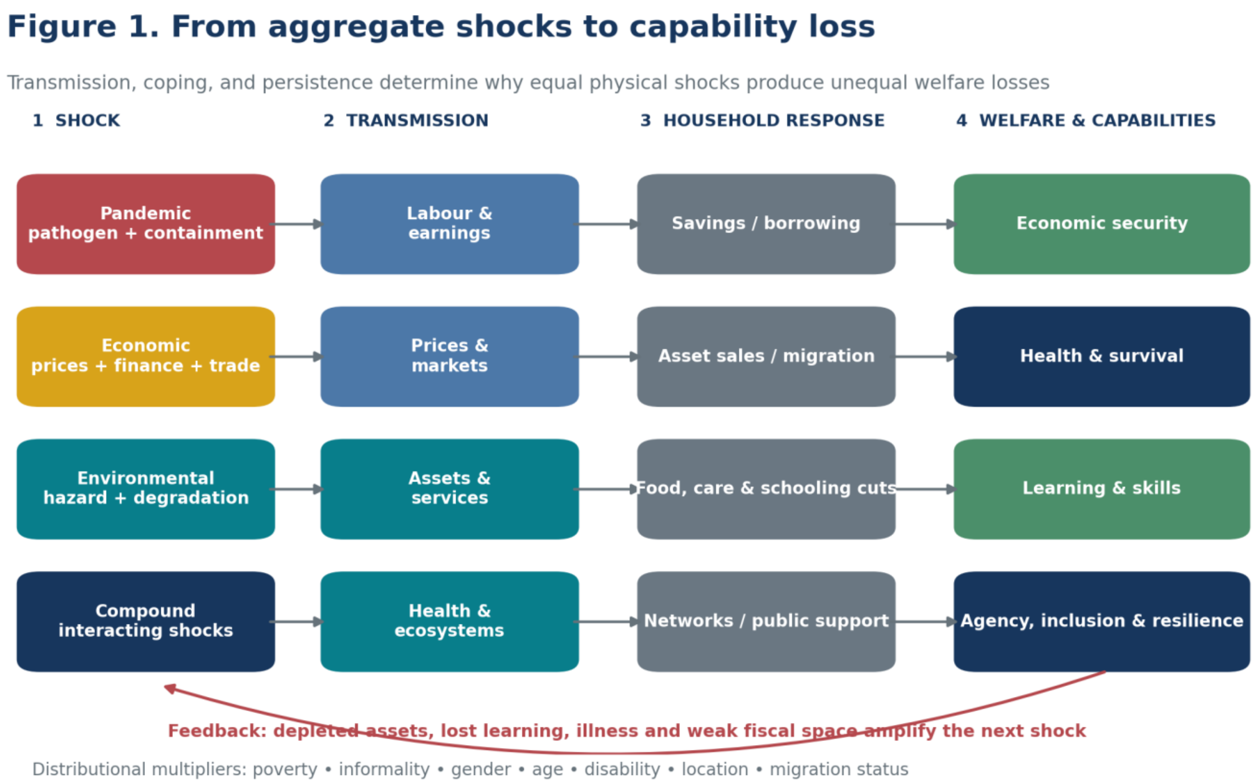

Figure 1. Conceptual pathways linking aggregate shocks to welfare losses.

Source: authors' synthesis of the study-level evidence register.

The framework highlights three reasons that distribution cannot be appended as an afterthought. Exposure differs: informal settlements, floodplains, heat-intensive workplaces, and tourism-dependent islands face higher physical or sectoral risk. Sensitivity differs: children, people with chronic illness, pregnant women, and older persons face distinct biological risks. Coping capacity differs: access to savings, credit, insurance, digital learning, cooling, health care, secure housing, remittances, and social protection determines whether a shock is smoothed or converted into irreversible loss. The same asset damage can therefore produce much larger consumption-equivalent loss for a poor household than for a wealthy one (Hallegatte et al., 2017).

1.2 Scope and contribution

The primary geographic scope is the developing member countries of the Asian Development Bank (ADB), grouped into East Asia, Southeast Asia, South Asia, Central and West Asia, and the Pacific. Regional estimates are preferred when their coverage is clear; evidence from Australia and other developed Asia-Pacific economies is used selectively where it identifies mechanisms or values health burdens relevant to the region. COVID-19 evidence begins in 2020. Economic shocks are restricted principally to shocks occurring from 2015 onward. Environmental evidence emphasizes the same period, but a small number of earlier exposure panels are retained when they provide unusually strong causal evidence about mechanisms, as with drought and rural consumption in the Philippines.

The review contributes a common extraction and appraisal structure across literatures that are usually separated. Each major study is recorded by geography, population, welfare indicator, quantitative magnitude, methodology, identification strategy, limitations, and confidence. The resulting synthesis is deliberately not a meta-analysis:

outcomes are too heterogeneous, and many headline estimates overlap. Instead, it offers a measurement architecture, an evidence map, and a disciplined comparative assessment.

2. Review method and interpretation

The search covered peer-reviewed economics, public health, climate, and interdisciplinary journals, with priority given to the outlets specified in the commission, and institutional evidence from ADB, the World Bank, IMF, ILO, WHO, UNICEF, UNDP, and ESCAP. Searches combined geographic terms with shock terms and welfare outcomes. Backward and forward citation chaining were used around influential studies. The evidence cutoff was 31 July 2026. A source was included when it provided a quantitative welfare estimate, a transparent model that could be interpreted as a welfare loss, or a high-quality post-disaster accounting of damage, economic loss, mortality, or poverty.

The evidence is classified into five designs. Causal and quasi-experimental studies use exogenous variation, discontinuities, instrumental variables, or panel weather variation. Repeated and high-frequency surveys measure changes during the shock but often lack an unexposed control. Excess-mortality and burden-of-disease studies construct epidemiological counterfactuals. Microsimulation and macroeconomic models compare observed or scenario outcomes with a no-shock baseline. Post-disaster needs assessments account for direct damage and disrupted economic flows. Each design answers a different question; their numerical outputs should not be pooled mechanically.

Confidence is rated high when the source is peer reviewed or an official statistical assessment, the population and counterfactual are transparent, the result is robust, and the estimate closely measures welfare. Medium confidence denotes credible but model-dependent estimates, incomplete representativeness, or indirect welfare proxies. Low confidence denotes severe data constraints, weak attribution, or strong dependence on untestable assumptions. Confidence refers to the estimate as used here, not to the overall quality of the organization or journal.

Several accounting rules govern comparison. GDP losses are flows relative to a counterfactual and are not added to asset damage. Poverty changes are not added across reports unless their lines, years, prices, and baselines match. Mortality totals are not summed across WHO and Lancet excess-mortality models. Lifetime earnings losses are present or undiscounted values depending on the source and remain projections. Exposure counts, such as people living in flood zones, are not treated as realized welfare losses. Climate projections are reported by scenario and horizon, and observed event losses are separated from climate-attributable fractions. These rules are conservative, but they prevent the most common form of double counting in flagship reports.

3. COVID-19: a synchronized multidimensional welfare shock

3.1 Output, employment, and poverty

The pandemic was the largest synchronized short-run shock in the review period. ADB estimated that developing Asia's GDP was 6.0-9.5% below a no-COVID counterfactual in 2020 and 3.6-6.3% below it in 2021 (Asian Development Bank, 2021a). These are not observed contractions: they are foregone output relative to the trajectory expected before the pandemic, an important distinction for economies that still posted positive growth. The regional nature of the loss reflects simultaneous domestic restrictions, global demand changes, tourism collapse, supply-chain interruptions, and precautionary behaviour.

Labour-market estimates show why GDP understates household harm. The ILO estimated that Asia and the Pacific lost 81 million jobs in 2020 and that lost working hours pushed an additional 22-25 million workers into working poverty (International Labour Organization, 2020). Informal workers were especially exposed because work could not be performed remotely and job separation rarely triggered unemployment insurance. In Southeast Asia, ADB estimated that 9.3 million jobs disappeared and 4.7 million additional people lived in extreme poverty in 2021 relative to the counterfactual (Asian Development Bank, 2022a). Across developing Asia, approximately 75-80 million more people lived in extreme poverty in 2020 than would have without COVID-19, with roughly 170-180 million additional people below a broader poverty threshold (Asian Development Bank, 2021b).

Household surveys reveal losses hidden by aggregates. Egger et al. (2021) harmonized evidence from more than 30,000 respondents in nine low- and middle-income countries, including Bangladesh, Nepal, and the Philippines. Across their 16 samples, the share reporting income decline ranged from 8% to 87%, with a median near 70%; employment declines ranged from 5% to 49%, with a median near 30%; and the share missing or reducing meals ranged from 9% to 87%, with a median near 45%. Only a minority received government assistance. Because these are before-after changes rather than causal treatment effects, they cannot isolate lockdowns from fear, infection, or global recession. Their strength is different: they document the simultaneity of labour, income, food, and market-access failures across diverse settings.

The burden was unequal within countries and across firm size. In developing East Asia and the Pacific, 32 million people were prevented from escaping poverty in 2021, while microenterprise sales fell by around one-third compared with roughly one-quarter among large firms (World Bank, 2021a). Harmonized phone-survey evidence from developing countries found that women were 8 percentage points more likely than men to stop working, young adults 4 points more likely than older adults, and less-educated workers 4 points more likely than their better-educated peers (Kugler et al., 2023). Sectoral composition explained only part of the gender gap, indicating that care responsibilities, norms, occupational status, and employer decisions also mattered.

3.2 Mortality, morbidity, and mental health

Mortality is the clearest intrinsic welfare loss, yet it is among the most contested pandemic quantities because civil-registration coverage is incomplete. The COVID-19 Excess Mortality Collaborators (2022) estimated 18.2 million global excess deaths during 2020-2021, compared with 5.94 million reported COVID-19 deaths. South Asia recorded the largest regional burden, including an estimated 4.07 million excess deaths in India, 736,000 in Indonesia, and 664,000 in Pakistan. WHO's independently constructed model produced a lower global total of 14.9 million, with 84% concentrated in South-East Asia, Europe, and the Americas (World Health Organization, 2022). The disagreement is substantive but does not overturn the core conclusion: official reported deaths substantially understated the pandemic mortality burden, especially where registration and testing were weak.

Age structured the direct health loss. O'Driscoll et al. (2021) estimated an infection fatality ratio of 0.001% at ages 5-9 and 8.29% at age 80 and above in the pre-vaccination period; within the oldest group, the estimate was 10.83% for men and 5.76% for women. These estimates are not a direct measure of Asia's realized deaths, and vaccines and variants changed the risk. They nevertheless explain why older persons bore such a disproportionate mortality loss and why interruptions to routine care, social contact, and family caregiving compounded the direct disease burden.

Mental-health losses were large but measured less precisely at country level. The COVID-19 Mental Disorders Collaborators (2021) estimated 53.2 million additional cases of major depressive disorder, a 27.6% increase, and 76.2 million additional anxiety-disorder cases, a 25.6% increase, in 2020. Females and younger people experienced larger increases. Country estimates were derived from a model linking available prevalence studies to infection and mobility indicators; direct surveys were sparse in many Asian settings. The results therefore establish a broad global burden and distributional gradient more securely than any precise national total.

Indirect mortality through disrupted services was especially consequential for women and children. UNICEF (2021) modelled approximately 228,000 additional child deaths and 11,000 additional maternal deaths in South Asia in 2020 under observed service-disruption scenarios. The estimate combines reductions in nutrition, immunization, antenatal, and obstetric services with epidemiological models. It should not be added to excess mortality without careful reconciliation, but it makes a crucial conceptual point: pandemic deaths include failures of ordinary health protection, not only infection deaths.

3.3 Learning, nutrition, and lifetime earnings

School closure converted a temporary emergency into a potentially lifelong distributional shock. In developing East Asia and the Pacific, students' average annual future earnings were projected to fall by about $524 in 2017 purchasing-power-parity terms, or 3.8%; the projected loss reached 9.3% in the ASEAN-5 and 7.9% in small East Asian economies (World Bank, 2021b). In Indonesia, a scenario of approximately half a year of lost learning, equivalent to 16 PISA reading points, implied $222.4 billion in lifetime income losses across 68 million students, equal to 19.9% of 2019 GDP (Yarrow et al., 2020). The GDP comparison communicates scale but should not be interpreted as a one-year output loss: it is a long-horizon earnings value for an entire cohort.

South Asia faced longer closures and weaker remote-learning access. Regional learning poverty was estimated to have increased from around 60% to 78%, and current students may lose up to 14.4% of future earnings (World Bank, 2023a). On average, 30 days of closure were associated with about 32 days of learning loss, suggesting that learning often regressed rather than merely paused. Poor children were less likely to engage with remote education; in East Asia and the Pacific the participation gap between children in the bottom two-fifths and others reached 20 percentage points in some settings (World Bank, 2021a). Devices, connectivity, parental time, home language, and school quality mediated the loss.

Nutrition shocks reinforced learning and health losses. Osendarp et al. (2021) linked pandemic income and service disruption to an estimated 9.3 million additional cases of child wasting, 2.6 million additional cases of stunting, 168,000 additional under-five deaths, and $29.7 billion in future productivity losses across 118 low- and middle-income countries during 2020-2022. South Asia accounted for a large share because of its high baseline burden. The model is scenario-based and global, but the pathway is empirically plausible: reduced household income, higher food prices, lower diet quality, and disrupted growth monitoring and treatment jointly damage early-life development.

3.4 Persistence and interpretation

The persistence of COVID-19 welfare loss varies by domain. Employment and sales recovered fastest where restrictions eased and demand returned, although debt, depleted working capital, and firm closure left scars. Poverty fell as growth resumed, but households that sold productive assets or accumulated high-cost debt remained less resilient. Mortality is irreversible. Untreated disease can generate chronic disability. Learning and stunting losses occur during sensitive periods, so later remediation must be unusually strong to restore the counterfactual trajectory. Mental health can improve as uncertainty declines, but bereavement, social isolation, violence, and economic insecurity may persist.

Table 1 summarizes the major COVID-19 studies. The strongest conclusion is not that any one estimate fully captures welfare. It is that every major domain moved adversely at once and that the losses interacted. Output contraction reduced fiscal space as health and social-protection needs surged; school closure increased care burdens and constrained women's employment; food insecurity damaged child development; and mortality removed earners and caregivers. COVID-19 therefore ranks first in regional breadth and multidimensional simultaneity, even though certain disasters and national crises produced larger proportional losses in particular countries.

Study	Geography	Population	Welfare indicator	Estimated loss	Methodology
Asian Development Bank (2021a)	Developing Asia	All residents	GDP relative to a no-pandemic baseline	6.0%-9.5% of regional GDP in 2020 and 3.6%-6.3% in 2021	Global macroeconomic scenario model with epidemiological and containment channels
Asian Development Bank (2021b)	Developing Asia	People near international poverty lines	Extreme and moderate poverty	75-80 million additional people in extreme poverty in 2020; roughly 170-180 million additional people below the broader poverty line	Growth-to-poverty simulation against a no-COVID baseline
International Labour Organization (2020)	Asia and the Pacific	Working-age population	Employment and working poverty	81 million jobs lost in 2020; working-hour losses pushed an additional 22-25 million workers into working poverty	ILO nowcasting model combining labour-force, national accounts, and high-frequency data
Egger et al. (2021)	Nine LMICs, including Bangladesh, Nepal, and the Philippines	More than 30,000 households across 16 samples	Income, employment, food security, and market access	Income fell in 8%-87% of samples (median about 70%); employment fell 5%-49% (median 30%); 9%-87% missed or reduced meals (median 45%)	Harmonized rapid phone surveys linked to pre-pandemic samples
World Bank (2021a)	Developing East Asia and Pacific	Households, children, and firms	Poverty, learning participation, and business sales	32 million people were prevented from escaping poverty; children in the bottom two-fifths were up to 20 percentage points less likely to be engaged in learning; microenterprise sales fell about one-third	Regional macro-poverty projections and harmonized high-frequency surveys
World Bank (2021b)	Developing East Asia and Pacific	Students	Lifetime earnings	Average annual earnings loss of about $524 per student in 2017 PPP, or 3.8%; estimated loss reaches 9.3% in ASEAN-5 and 7.9% in small East Asian economies	Learning-adjusted schooling losses mapped to returns to education and lifetime earnings
Yarrow et al. (2020)	Indonesia	About 68 million students	Learning and lifetime income	Roughly 0.5 year of learning or 16 PISA reading points lost; $222.4 billion in lifetime income, equivalent to 19.9% of 2019 GDP	School-closure scenarios linked to learning-adjusted years of schooling and returns to education
World Bank (2023a)	South Asia	Children and young people	Learning poverty and future earnings	Learning poverty rose from about 60% to 78%; current students may lose up to 14.4% of future earnings; every 30 days of closure was associated with about 32 days of learning loss	Regional synthesis of assessment data, closure duration, and human-capital simulations
Osendarp et al. (2021)	118 low- and middle-income countries, with heavy burden in South Asia	Children under five and women	Wasting, stunting, maternal nutrition, deaths, and productivity	Moderate 2020-2022 scenario: 9.3 million additional wasted children, 2.6 million additional stunted children, 168,000 additional under-five deaths, and $29.7 billion in future productivity losses	Linked MIRAGRODEP, nutrition-risk, LiST, and Optima Nutrition models
UNICEF (2021)	South Asia	Women and children	Maternal and child mortality	Service disruptions may have contributed to about 228,000 additional child deaths and 11,000 additional maternal deaths in 2020	Country service-disruption scenarios combined with LiST-type mortality modelling

Study	Geography	Population	Welfare indicator	Estimated loss	Methodology
COVID-19 Mental Disorders Collaborators (2021)	204 countries and territories, including Asia-Pacific	All ages; sex- and age-disaggregated	Major depressive and anxiety disorders and DALYs	53.2 million additional major-depressive-disorder cases (+27.6%) and 76.2 million additional anxiety cases (+25.6%) in 2020; females and younger people bore larger increases	Systematic review and Bayesian meta-regression linked to mobility and infection indicators
COVID-19 Excess Mortality Collaborators (2022)	191 countries and territories	All ages	Excess deaths	18.2 million global excess deaths in 2020-2021 versus 5.94 million reported COVID deaths; India 4.07 million, Indonesia 736,000, Pakistan 664,000	All-cause mortality comparison plus ensemble models for data-poor locations
Kugler et al. (2023)	40 developing countries, including East and South Asia	Working-age adults	Work stoppage by sex, age, education, and location	Women were 8 percentage points more likely than men to stop work; youth 4 points more likely than older adults; low-education and urban workers each about 4 and 3 points more likely	Harmonized high-frequency phone surveys with multivariate decomposition
O'Driscoll et al. (2021)	45 countries with 22 seroprevalence studies	All ages	Infection fatality ratio by age and sex	IFR rose from 0.001% at ages 5-9 to 8.29% at age 80+; among those 80+, estimated IFR was 10.83% for men and 5.76% for women	Bayesian ensemble model combining age-specific deaths and seroprevalence

Table 1. Major studies on COVID-19 welfare losses in Asia and the Pacific. Source: authors' synthesis of cited studies.

4. Economic shocks since 2015

4.1 Inflation and the cost of living

Inflation is a welfare shock when nominal resources fail to keep pace with the prices of the goods that households actually consume. The headline consumer price index can misstate this burden because poor households devote larger shares to food, cooking fuel, transport, and rent; price changes also differ across locations and product quality. Static budget-incidence studies provide a useful first-order loss but omit substitution, own production, wage adjustment, producer income, and policy responses.

The 2022 food and energy surge reduced purchasing power across developing East Asia and the Pacific by amounts ranging from approximately 2% in Viet Nam to 11% in Mongolia in the country cases examined by the World Bank (2022a). IMF microsimulations estimated that projected food and energy inflation increased relative poverty by around 1 percentage point in Cambodia and Viet Nam and 0.2 point in China; a more severe combined price shock approximately doubled the effect (International Monetary Fund, 2022). These estimates are smaller than the pandemic's poverty shock at regional scale, but the distribution is more concentrated because food-price exposure rises as income falls.

The nutrition literature shows why a temporary price spike may have a permanent component. Using 1.27 million children in 44 low- and middle-income countries, Headey and Ruel (2023) found that a 5% increase in real food prices during the preceding three months was associated with a 9% increase in the risk of wasting and a 14% increase in severe wasting. For asset-poor children, the wasting risk rose by 15%, compared with 6% for non-poor children. Infants were especially vulnerable, and prenatal and first-year food-price exposure predicted later stunting. These are relative-risk changes rather than percentage-point increases, but they supply unusually strong micro-level evidence connecting inflation to biological human-capital loss.

The UNDP's broader simulation across 159 developing countries estimated that the 2022 price surge could push 71 million people into poverty in only three months, with Armenia, Pakistan, Sri Lanka, and Uzbekistan among the Asian hotspots (United Nations Development Programme, 2022a). That global total is not an Asia-Pacific estimate and should not be added to IMF or World Bank results. Its policy comparison is nevertheless instructive: targeted transfers protected more poor people per dollar than universal energy subsidies, because generalized subsidies transfer large benefits to higher-consuming households and can crowd out health, education, and infrastructure.

4.2 Debt, currency, and macroeconomic crises

National crises generated some of the largest non-pandemic welfare reversals. Sri Lanka's poverty rate rose from 13.1% in 2021 to 25.0% in 2022, implying roughly 2.5 million additional poor people; an estimated half a million industry and services jobs were lost, and annual average inflation reached 46.4% in 2022 (World Bank, 2023b). These outcomes reflect an interacting balance-of-payments crisis, debt distress, currency depreciation, shortages, fiscal adjustment, policy failures, global commodity prices, and residual pandemic damage. Because there is no credible single-shock counterfactual, the evidence identifies the magnitude of the crisis-era welfare setback more securely than the contribution of each cause.

Lao PDR illustrates how exchange-rate and debt pressure reaches households. With consumer inflation near 39% in 2023 and average nominal wages rising only 5.7%, measured real wages fell by roughly one-third (World Bank, 2024a). In 2024, low-income real per-capita income declined by 6.9%; more than one-third of low-income households faced moderate or severe food insecurity, and sizeable shares reduced education and health spending. Such coping protects calories or debt service in the short run at the cost of future productivity and health.

In Mongolia, the commodity and macroeconomic slowdown raised poverty from 21.6% in 2014 to 29.6% in 2016, with a 10.1 percentage-point rise in rural areas (World Bank, 2017). In Afghanistan, the political transition, aid freeze, banking disruption, drought, and institutional breakdown produced an estimated 20% GDP contraction during 2020-2021 and a one-year output loss near $5 billion; excluding women from employment was estimated to cost up to $1 billion, or 5% of GDP (United Nations Development Programme, 2022b). In Myanmar, poverty was estimated at 32.1% in 2024, with approximately 7 million more poor people than before COVID-19 and GDP about 9% below its 2019 level (World Bank, 2024b). The Myanmar estimate receives lower confidence because conflict and restricted data collection weaken representativeness and attribution.

These cases demonstrate a cumulative mechanism. Depreciation raises food, fuel, fertilizer, and medicine prices; monetary tightening and banking stress reduce credit; fiscal consolidation lowers transfers and services; and recession reduces wages and employment. Poor households then cut diet quality, schooling, preventive health care, and productive investment. A crisis can therefore improve headline indicators during partial stabilization while welfare remains below the pre-shock path because assets, human capital, and institutional capacity have already deteriorated.

4.3 Financial, monetary, trade, and supply-chain disruptions

Quasi-experimental evidence is rare for macro shocks, but India's 2016 demonetization provides a strong case. Chodorow-Reich et al. (2020) used cross-district differences in dependence on withdrawn currency denominations as an instrument for cash shortage. Employment and night-light output fell by at least 2 percentage points in the fourth quarter of 2016, while bank credit fell by about 2 points; the effects dissipated over subsequent months. The study establishes that nominal liquidity disruptions can produce real losses even without a conventional destruction of productive capacity, particularly where cash supports informal exchange.

Trade and geopolitical shocks are generally evaluated through models rather than observed household counterfactuals. Abiad et al. (2018) estimated that the tariffs implemented during the early United States-China trade conflict would lower China's GDP by around 1% over two to three years and US GDP by about 0.2%, with mixed trade-diversion gains elsewhere in developing Asia. An IMF scenario placed the long-run global GDP loss near 0.4%, or $340 billion in 2018 dollars, and the effect on China's 2019 GDP level near 1.6% (International Monetary Fund, 2019). These results show the order of magnitude of output exposure but not the welfare incidence. Export workers, import consumers, firms using intermediate inputs, and economies gaining redirected investment can experience opposite effects.

Supply-chain disruption is difficult to isolate because it was intertwined with pandemic closures, shipping congestion, semiconductor shortages, commodity prices, and changes in demand. Most regional estimates quantify production or trade rather than household welfare. This is an important evidence gap rather than evidence of a small effect. The welfare pathway is clear—input shortages reduce hours and output, delays raise prices, and inventories and working capital are depleted—but credible estimates that trace these changes into consumption, poverty, and health in ADB developing members remain limited.

Table 2 summarizes the major economic-shock evidence. Confidence is highest for household surveys and quasi-experimental designs that measure realized changes, moderate for transparent microsimulations, and lower where conflict or data collapse prevents representative measurement. Across the category, food inflation and currency crises are especially damaging to poor households because they combine a high expenditure exposure with few substitution and smoothing options.

Study	Shock type	Geography	Welfare measure	Magnitude
Headey and Ruel (2023)	Real food-price inflation	44 low- and middle-income countries, including South and Southeast Asia	Wasting and stunting	A 5% rise in real food prices over the previous three months increased wasting risk 9% and severe wasting 14%; the increase was 15% for asset-poor children and 6% for non-poor children
International Monetary Fund (2022)	Food and energy inflation	Selected Asian emerging and developing economies	Real income and relative poverty	Projected 2022 inflation raised relative poverty about 1 percentage point in Cambodia and Viet Nam and 0.2 point in China; a severe combined price shock roughly doubled the effects
World Bank (2022a)	Cost-of-living inflation	Developing East Asia and Pacific	Purchasing power	Inflation eroded household purchasing power by about 2% in Viet Nam and 11% in Mongolia across the country cases examined
United Nations Development Programme (2022a)	Food and energy price surge	159 developing countries	Poverty	About 71 million people could be pushed into poverty within three months; Asian hotspots included Armenia, Pakistan, Sri Lanka, and Uzbekistan
World Bank (2023b)	Debt, balance-of-payments, inflation, and recession crisis	Sri Lanka	Poverty, jobs, living costs, and nutrition	Poverty rose from 13.1% in 2021 to 25.0% in 2022 at the $3.65 2017 PPP line, adding about 2.5 million poor people; an estimated half a million industry and services jobs were lost, with annual average inflation of 46.4% in 2022
World Bank (2024a)	Currency depreciation and high inflation	Lao PDR	Real wages, consumption, food security, health, and education	With 39% inflation in 2023 and nominal wages up 5.7%, real wages fell about 33%; in 2024 low-income real per-capita income fell 6.9%, and 36.2% of low-income households faced moderate or severe food insecurity
World Bank (2017)	2015-2016 commodity and macroeconomic slowdown	Mongolia	Poverty	National poverty rose from 21.6% in 2014 to 29.6% in 2016; rural poverty increased by 10.1 percentage points
United Nations Development Programme (2022b)	Political transition, aid and banking freeze, and macroeconomic collapse	Afghanistan	GDP, poverty, food prices, and female employment	GDP contracted about 20% in 2020-2021, a one-year loss near $5 billion; the basic food basket rose about 35%; excluding women from work could cost up to $1 billion or 5% of GDP
World Bank (2024b)	Conflict, political disruption, inflation, and economic contraction	Myanmar	Poverty and economic insecurity	Poverty reached 32.1%, with about 7 million more poor people than before COVID-19; GDP remained roughly 9% below its 2019 level
Chodorow-Reich et al. (2020)	2016 demonetization and cash shortage	India	Employment, night-light output, and credit	District-level cash shortages reduced employment and night-light output by at least 2 percentage points in 2016 Q4 and bank credit by about 2 points; effects dissipated over subsequent months
Abiad et al. (2018)	Tariffs and trade conflict	Developing Asia with China-US trade conflict	GDP and employment reallocation	Tariffs implemented at the time were estimated to reduce China's GDP by about 1% over two to three years and US GDP by about 0.2%, with mixed trade-diversion gains elsewhere in developing Asia
International Monetary Fund (2019)	US-China trade conflict	Global, China, and United States	Long-run GDP	The tariff conflict was estimated to lower long-run global GDP by about 0.4% ($340 billion in 2018 dollars); China's 2019 GDP level by about 1.6% and US GDP by about 1%

Table 2. Major studies on economic shocks, 2015-present. Source: authors' synthesis of cited studies.

5. Environmental degradation, disasters, and climate change

5.1 From asset damage to welfare loss

Disaster accounting conventionally distinguishes damage to physical assets from losses in economic flows. Neither equals welfare. The same destroyed dwelling imposes a larger utility and consumption loss on a household without savings, insurance, secure land tenure, or public support. Hallegatte et al. (2017) formalized this distinction for 117 countries. Their model estimated annual global consumption-equivalent wellbeing losses of $520 billion—substantially above direct asset losses—and about 26 million people pushed into extreme poverty each year. The estimate is global rather than an Asia-Pacific subtotal, but its conceptual contribution is central: socioeconomic resilience governs the welfare multiplier applied to physical exposure.

This framework also explains why repeated small events can matter more than a single recorded catastrophe. Families may rebuild a roof after each flood while selling livestock, withdrawing children from school, or accumulating debt. National accounts can recover once reconstruction begins, yet the household's asset position, security, and future earnings remain lower. Disaster loss databases capture major events better than recurrent urban flooding, localized landslides, salinization, and heat stress; the measured burden is therefore biased toward visible assets and away from informal livelihoods and care.

5.2 Floods, earthquakes, cyclones, and compound disasters

Flood risk is concentrated in Asia. Rentschler, Salhab, and Jafino (2022) estimated that 1.81 billion people worldwide are exposed to a 1-in-100-year flood, of whom 1.24 billion live in South and East Asia. China and India each account for roughly 390-395 million exposed people, and 170 million exposed people globally live in extreme poverty. The study measures exposure, not realized loss, but its spatial overlap of hazard and poverty identifies where a flood is most likely to become a welfare crisis. Tellman et al. (2021), using satellite images of 913 large floods, estimated that 255-290 million people were directly affected between 2000 and 2018 and that the proportion of the population in observed flood footprints grew 20-24% from 2000 to 2015. South and Southeast Asia recorded some of the largest increases.

The 2022 Pakistan floods show how exposure translates into multidimensional loss. The government-led post-disaster needs assessment estimated $14.9 billion in direct damage and $15.2 billion in economic losses. Thirty-three million people were affected, 1,730 died, and poverty was projected to rise by 3.7-4.0 percentage points, pushing 8.4-9.1 million additional people below the poverty line (Government of Pakistan et al., 2022). The largest damage occurred in housing, agriculture, livestock, and transport—sectors closely tied to poor households' assets and food security. Health effects continued through stagnant water, disease, interrupted services, and displacement, so even the large PDNA totals are incomplete welfare measures.

The 2015 Nepal earthquakes caused over 8,790 casualties and 22,300 injuries, affected the lives of about 8 million people, and generated total recovery needs of US$6.7 billion; poverty was estimated to increase by 2.5-3.5 percentage points, translating into at least 700,000 additional poor (Government of Nepal, 2015). Housing accounted for a dominant share, with rural and poor households disproportionately affected. Because trade disruption followed the earthquakes, the later macro outcome cannot be attributed to seismic damage alone. The assessment nevertheless demonstrates how a physical event can reverse years of poverty reduction through housing loss, livelihood disruption, injury, and debt.

Pacific island economies display the largest event losses relative to national income. Tropical Cyclone Pam caused damage and losses near $450 million in Vanuatu, equivalent to 64% of GDP, displaced about 65,000 people, and compromised the livelihoods of at least 80% of the rural population (Government of Vanuatu, 2015). Tropical Cyclone Winston caused 44 deaths and approximately $1.38 billion in damage and losses in Fiji, equivalent to 31% of GDP; more than 30,000 houses were damaged or destroyed and around 62% of the population was affected (Government of Fiji, 2016). The 2022 Hunga Tonga-Hunga Ha'apai eruption, tsunami, and ashfall caused direct damage estimated at $90.4 million, or 18.5% of Tonga's GDP, excluding indirect losses and uncertain ashfall costs (World Bank, 2022b). These ratios should not be added to annual GDP contraction, but they show how a single event can overwhelm domestic fiscal and insurance capacity.

Coping capacity is itself stratified by wealth. Giannelli and Canessa (2021) mapped the 2014 Bangladesh flood to a household panel and found that internal migration rose seven percentage points among low-wealth households while international migration rose three points among high-wealth households. Remittances from established international migrants offset roughly 40% of the decline in farm self-employment income and half the decline in food expenditure for poorer households. The averages conceal the mechanism: the poorest substitute the cheaper coping strategy for the one they cannot afford.

5.3 Heat, labour productivity, and mortality

Heat reduces welfare through mortality, illness, lost work capacity, school disruption, sleep, and higher cooling expenditure. Somanathan et al. (2021) combined daily worker records from selected Indian firms with a national panel of manufacturing plants. Productivity declined and absenteeism increased on hot days, while climate control mitigated the loss; annual plant output fell about 2% per additional degree Celsius. The design uses within-location weather variation and directly observes production, making it one of the strongest estimates of a labour-productivity mechanism in developing Asia. It covers formal manufacturing better than agriculture, construction, transport, and informal outdoor work, where cooling and labour protection are often weaker.

Heat mortality is both observed and projected. Vicedo-Cabrera et al. (2021) attributed approximately 37% of warm-season heat-related deaths across 732 locations to anthropogenic climate change during 1991-2018. Among the nine included Asian countries, attributable shares ranged from around 21.3% in China to 67.7% in Kuwait. The city network is not nationally representative, but it provides direct evidence that climate change is already contributing to mortality, not merely posing a future risk.

Carleton et al. (2022) estimated future age-specific mortality-temperature relationships at subnational scale while modelling adaptation. Under a high-emissions pathway, the net global mortality cost of warming was valued at about 3.2% of GDP in 2100, and the mortality component of the social cost of carbon at $36.6 per tonne of carbon dioxide. This is a monetized partial welfare measure, not a forecast of total climate damage. It excludes morbidity, productivity, ecosystems, conflict, and direct disaster loss, while depending on income-sensitive valuation and assumptions about costly adaptation.

Physical climate models identify limits to adaptation. Im, Pal, and Eltahir (2017) projected that, under business-as-usual emissions, parts of the densely populated Indus and Ganges basins could approach or exceed a 35 degrees C wet-bulb threshold late in the century. Approximately 4% of South Asia's population could face such exceedances, and about three-quarters could face dangerous wet-bulb temperatures above 31 degrees C. These are exposure estimates, not deaths; realized welfare depends on work patterns, cooling, water, health, migration, and warning systems. Their importance lies in identifying nonlinear physiological constraints that conventional GDP models may miss.

Heat also kills before birth confers any protection. Banerjee and Maharaj (2019) found that high temperature during pregnancy raises infant mortality by about two deaths per 1,000 births in India, a relationship that holds in rural areas only and is plausibly mediated by reduced agricultural yields and wages and by greater diarrhoeal disease. Their adaptation test is the more policy-relevant result: a phased employment-guarantee programme did not modify the temperature-mortality relationship, while a community health-worker programme did. Income support and health delivery are not interchangeable responses to heat.

5.4 Drought, food systems, and child development

Drought damages welfare through crop and livestock loss, food prices, water collection, disease, migration, and conflict over resources. In rural Philippines data, drought reduced food consumption by approximately 4% (Skoufias et al., 2013). Although the underlying events predate 2015, the result is retained because it directly links rainfall variation to a household welfare outcome. The inability to fully smooth food consumption indicates that credit, savings, remittances, and public transfers were insufficient.

Early-life exposure can convert a weather shock into a lifelong loss. Groppo and Kraehnert (2016) used cohort and spatial variation in Mongolia's 2009-2010 dzud, during which 10.3 million livestock died. Children from herding households exposed in utero or infancy in severely affected districts had substantially lower height-for-age, with a preferred interaction estimate near 1.67 standard deviations. The sample is limited, and the event was exceptional, but the panel design supports a causal interpretation. For pastoral households, livestock are simultaneously an income source, asset, food source, and buffer; their destruction collapses several capability supports at once.

Food prices connect economic and environmental shocks. Drought, heat, flood, energy costs, fertilizer shortages, trade restrictions, and currency depreciation can all raise local food prices. The Headey and Ruel (2023) result—that a 5% real food-price increase raises child wasting risk 9%—therefore represents a common final pathway rather than a shock-specific coefficient. Treating food and nutrition systems as resilience infrastructure is consequently justified across the economic and climate portfolios.

Rainfall shocks work through the same seasonal income channel in the opposite direction. Tiwari, Jacoby, and Skoufias (2016) exploited monsoon timing in rural Nepal and found that a 10% rainfall surplus above historic norms raised weight-for-height by 0.13 standard deviations among children under five, decomposing into an income effect as large as 0.17 standard deviations against a disease-environment effect of at most 0.04. The corresponding height gain proved transitory, disappearing by age five, which is a useful caution: a measurable anthropometric response to a weather shock is not by itself evidence of a permanent human-capital loss.

5.5 Wildfire, air pollution, and environmental degradation

Environmental degradation produces chronic welfare loss even without a discrete disaster declaration. The India State-Level Disease Burden Initiative Air Pollution Collaborators (2021) attributed 1.67 million deaths in 2019, 17.8% of all deaths, to ambient and household air pollution. Lost output from premature mortality and morbidity was valued at $36.8 billion, or 1.36% of GDP. This is narrower than willingness-to-pay welfare valuation and excludes pain, caregiving, and non-market activity, so it should be read as an economic lower bound rather than a complete social cost.

Quasi-experimental evidence strengthens the causal case. China's Huai River policy created a sharp difference in coal heating and particulate exposure. Ebenstein et al. (2017) estimated that a 10 microgram per cubic metre increase in PM10 reduced life expectancy by 0.64 year; national compliance with China's Class I standard would save an estimated 3.7 billion life-years. The pollutant mix and historical setting differ from current PM2.5 exposure, but the design shows that sustained pollution produces exceptionally large longevity losses.

Fire creates acute peaks within the chronic pollution burden. Koplitz et al. (2016) modelled approximately 100,300 premature deaths across Indonesia, Malaysia, and Singapore from the 2015 equatorial Asian haze, more than 90,000 in Indonesia. The result is model-based and far exceeds contemporaneous reported acute deaths because it applies concentration-response functions to population exposure; emissions and atmospheric transport introduce uncertainty. In a higher-income comparator, Johnston et al. (2021) attributed 417 excess deaths, more than 3,100 cardiorespiratory hospital admissions, 1,305 asthma emergency visits, and A$1.95 billion in health costs to smoke from Australia's 2019-2020 megafires. Even this valuation excludes mental health, ecosystem loss, displacement, housing, and long-term disease.

5.6 Long-run climate damage

Long-run climate estimates are large but should be presented as conditional scenarios. ADB's Asia-Pacific Climate Report estimated that regional GDP could be 17% lower by 2070 and 41% lower by 2100 under a high-end emissions pathway; a Paris-aligned pathway limited the 2100 loss to around 11% (Asian Development Bank, 2024). Sea-level rise and labour-productivity loss were dominant channels, and up to 300 million people could be exposed to coastal inundation risk. ESCAP (2023) estimated annualized disaster losses of $953 billion, or 3.0% of regional GDP, at 1.5 degrees C warming and nearly $1 trillion at 2 degrees C.

Reduced-form global studies support the direction while disagreeing on magnitude. Burke, Hsiang, and Miguel (2015) estimated that unmitigated warming would make global income approximately 23% lower in 2100 than without climate change. Diffenbaugh and Burke (2019) attributed a 25% increase in population-weighted between-country inequality to historical warming, and found per-capita GDP reduced by 17%-31% across the poorest four deciles of the population-weighted country distribution, with a median impact of about -27% among the countries whose cumulative per-capita emissions are lowest. Both results rely on a nonlinear temperature-growth relationship accumulated over time. Critics question extrapolation, adaptation, and the persistence of annual growth effects. The estimates are best interpreted as evidence of potentially large macroeconomic risk and unequal historical burden, not point forecasts.

Country models give policy-relevant ranges. Without adaptation, Viet Nam's GDP could be 12-14.5% lower by 2050, with cumulative losses of $400-$523 billion; tropical cyclones already impose losses around 0.8% of GDP annually (World Bank, 2025). In India, a carbon-intensive scenario could reduce GDP by 2.8% and depress living standards for nearly half the population by 2050 (Mani et al., 2018). Across these models, adaptation assumptions are decisive. The correct inference is not that one number is definitive, but that delay locks in capital, settlement, health, and labour systems that amplify future loss.

Table 3 separates observed events, causal micro estimates, exposure measures, burden-of-disease attribution, and long-run scenarios. This separation is necessary to avoid the false precision of a single environmental-loss total.

Study	Hazard type	Geography	Welfare outcome	Magnitude of loss
Hallegatte et al. (2017)	Natural disasters	117 countries, including Asia-Pacific	Consumption-equivalent wellbeing and poverty	$520 billion in annual global consumption-equivalent wellbeing losses and 26 million people pushed into extreme poverty each year; resilience policies could cut wellbeing losses about 20%
Asian Development Bank (2024)	Long-run climate change	Developing Asia and the Pacific	GDP, labour productivity, coastal assets, and poverty risk	Under a high-end emissions pathway, regional GDP could be 17% lower by 2070 and 41% lower by 2100; a Paris-aligned pathway limits the 2100 loss to around 11%; up to 300 million people face coastal-inundation risk
ESCAP (2023)	Multi-hazard disasters under warming	Asia and the Pacific	Annualized disaster losses	At 1.5 degrees C warming, annualized losses could reach $953 billion or 3.0% of regional GDP; at 2 degrees C they approach $1 trillion or 3.1%

Study	Hazard type	Geography	Welfare outcome	Magnitude of loss
World Bank (2025)	Climate change and tropical cyclones	Viet Nam and Pacific atoll economies	GDP and disaster damage	Without adaptation, Viet Nam's GDP could be 12%-14.5% lower by 2050, with $400-$523 billion cumulative losses; current cyclone losses average about 0.8% of GDP annually; in atoll economies annual disaster losses already reach 3%-7% of GDP
Burke, Hsiang, and Miguel (2015)	Long-run warming	Global panel with Asia-Pacific countries	Income per capita	Unmitigated warming was estimated to make global income about 23% lower in 2100 than in a world without climate change
Diffenbaugh and Burke (2019)	Historical anthropogenic warming, 1961-2010	Global, with country results for Asia	GDP per capita and between-country inequality	Warming increased population-weighted between-country inequality by about 25%; per-capita GDP was reduced 17%-31% at the poorest four deciles of the population-weighted country distribution, and the 18 countries with the lowest cumulative per-capita emissions had a median impact of about -27% relative to a no-warming counterfactual
Carleton et al. (2022)	Extreme heat under climate change	Global, including India and Asia-Pacific	Mortality and monetized mortality risk	Under high emissions, net mortality costs including adaptation were valued at about 3.2% of global GDP in 2100; the mortality partial social cost of carbon was $36.6 per tCO2
Vicedo-Cabrera et al. (2021)	Observed anthropogenic warming	732 locations in 43 countries, including nine Asian countries	Heat-related mortality	About 37% of warm-season heat-related deaths were attributable to anthropogenic climate change in 1991-2018; estimates across included Asian countries ranged roughly from 21.3% in China to 67.7% in Kuwait
Somanathan et al. (2021)	High temperatures	India	Worker productivity, absenteeism, and plant output	Annual plant output fell about 2% per 1 degree C increase; daily microdata show lower worker productivity and greater absenteeism, with climate control mitigating losses
Rentschler, Salhab, and Jafino (2022)	Fluvial, pluvial, and coastal flood risk	188 countries	Flood exposure by poverty status	1.81 billion people were directly exposed globally; 1.24 billion lived in South and East Asia, including 395 million in China and 390 million in India; 170 million exposed people lived in extreme poverty
Tellman et al. (2021)	Floods, 2000-2018	913 large floods in 169 countries	Population directly affected and growth of exposure	Between 255 million and 290 million people were directly affected; the share of global population in observed flood footprints grew about 20%-24% from 2000 to 2015
Government of Pakistan et al. (2022)	2022 monsoon floods	Pakistan	Damage, economic losses, poverty, and multidimensional poverty	$14.9 billion in damage and $15.2 billion in economic losses; poverty projected to rise 3.7-4.0 percentage points, adding 8.4-9.1 million poor people; 1,730 deaths
Government of Nepal (2015)	2015 earthquakes	Nepal	Deaths, damage, economic loss, recovery need, and poverty	Over 8,790 casualties and 22,300 injuries, with the lives of about 8 million people affected; total recovery needs of US$6.7 billion; World Bank simulations project an additional 2.5-3.5 percent of Nepalis pushed into poverty in FY 2015-2016, translating into at least 700,000 additional poor
Government of Vanuatu (2015)	Tropical Cyclone Pam	Vanuatu	Damage and economic loss	About $450 million in damage and losses, equivalent to 64% of GDP; roughly 65,000 people displaced and livelihoods of at least 80% of the rural population compromised
Government of Fiji (2016)	Tropical Cyclone Winston	Fiji	Deaths, damage and economic loss, housing	44 deaths and about $1.38 billion in damage and losses, equivalent to 31% of GDP; 30,369 houses were damaged or destroyed and about 62% of the population was affected
World Bank (2022b)	Volcanic eruption, tsunami, and ashfall	Tonga	Direct physical damage	$90.4 million in direct damage, equivalent to about 18.5% of GDP
Groppo and Kraehnert (2016)	2009-2010 dzud	Mongolia	Child height-for-age	Children exposed in utero or infancy in severely affected districts had height-for-age z-scores about 1.67 standard deviations lower in the preferred interaction estimate; 10.3 million livestock died
India State-Level Disease Burden Initiative Air Pollution Collaborators (2021)	Ambient and household air pollution	India	Deaths, disease burden, and output loss	1.67 million deaths in 2019 (17.8% of all deaths) were attributable to air pollution; output losses were $36.8 billion, or 1.36% of GDP
Ebenstein et al. (2017)	Long-term particulate pollution induced by the Huai River heating policy	China	Life expectancy	A 10 microgram/m3 increase in PM10 reduced life expectancy by about 0.64 year; meeting China's Class I standard was estimated to save 3.7 billion life-years
Koplitz et al. (2016)	2015 peat and forest-fire haze	Indonesia, Malaysia, and Singapore	Premature mortality from PM2.5 exposure	About 100,300 premature deaths were estimated across the three countries, more than 90,000 in Indonesia
Johnston et al. (2021)	Black Summer megafire smoke	Eastern Australia	Premature deaths, hospital use, and health costs	Smoke exposure was associated with 417 excess deaths, 1,124 cardiovascular and 2,027 respiratory hospital admissions, 1,305 asthma emergency visits, and A$1.95 billion in health costs

Table 3. Major studies on environmental and climate shocks. Exposure and scenario estimates are labelled and are not treated as realized loss.

6. Welfare losses over the life cycle

6.1 Children aged 0-17

Children experience smaller direct COVID-19 mortality risk than older adults but the longest potential duration of indirect loss. The main pathways are prenatal and early-life nutrition, interruption of preventive health care, school closure, household income loss, parental stress, displacement, and reduced protection from violence or exploitation. The evidence is strongest for learning and nutrition. South Asian learning poverty rose to an estimated 78%, and current students may lose up to 14.4% of future earnings (World Bank, 2023a). Indonesia's projected lifetime earnings loss reached $222.4 billion across a large cohort (Yarrow et al., 2020). These projections are sensitive to remediation, but even substantial catch-up does not automatically restore the pre-shock trajectory because learning is cumulative.

Nutrition losses begin earlier and can be less reversible. Pandemic service and income disruptions generated millions of modelled additional wasting and stunting cases (Osendarp et al., 2021). A 5% rise in real food prices raised wasting risk by 9%, with a larger effect among asset-poor infants (Headey and Ruel, 2023). The Mongolian dzud study shows that weather exposure in utero or infancy can materially reduce height-for-age (Groppo and Kraehnert, 2016). These effects combine biological sensitivity with household coping: adult calorie reduction, asset sales, and borrowing may be exhausted before a shock ends, leaving children's diets and care exposed.

The geographic concentration differs by mechanism. South Asia bears the largest aggregate burden of learning poverty, wasting, and pandemic health-service disruption. Southeast Asia combines school and labour-market disruption with high flood, typhoon, haze, and food-price exposure. Pastoral areas of Central and East Asia face livestock and winter shocks. Pacific children face repeated school and housing disruption from cyclones and other hazards, although quantitative learning estimates are sparse. The central evidence gap is longitudinal: few studies follow affected children into later schooling, labour-market entry, fertility, or adult health.

6.2 Working-age adults aged 18-64

Working-age adults are the primary transmission point from aggregate shocks to household income. During COVID-19, 81 million jobs were lost regionally, and income decline was reported by a majority in many rapid-survey samples (International Labour Organization, 2020; Egger et al., 2021). Informal workers lacked remote-work options and contributory protection; migrants faced border closure, job loss, and costly return; women absorbed increased unpaid care and had higher work-stoppage rates; and young adults experienced disrupted school-to-work transitions (Kugler et al., 2023).

Economic shocks act through real wages, unemployment, working capital, and migration. Lao workers experienced a measured real-wage decline near one-third during the high-inflation phase, while Sri Lanka lost approximately half a million jobs (World Bank, 2023b, 2024a). Demonetization in India reduced local employment and night-light output even though productive assets remained intact (Chodorow-Reich et al., 2020). These losses can persist through occupational downgrading, business closure, debt, and lower contributions to pensions and insurance.

Environmental risks are occupationally differentiated. Heat lowers factory productivity and attendance, but outdoor and informal workers may face greater physiological exposure and less cooling (Somanathan et al., 2021). Farmers and herders lose both current earnings and productive assets during drought, flood, cyclone, or dzud. Displacement breaks customer, employer, and care networks. Air pollution and wildfire smoke reduce healthy work time and raise medical expenditure. Mental distress is both an intrinsic loss and a mediator of labour supply, family functioning, and coping.

6.3 Older persons aged 65 and above

Older persons bore the largest direct mortality burden from COVID-19. The pre-vaccination infection fatality ratio increased steeply with age, reaching 8.29% for those aged 80 and above (O'Driscoll et al., 2021). Excess mortality also includes deaths arising from delayed treatment, overloaded health systems, and isolation. Welfare loss extends beyond survival: mobility restrictions reduced social contact, caregivers were unavailable or exposed to infection, chronic-disease monitoring was interrupted, and digital delivery of cash or health information could exclude people with limited connectivity or documentation.

Economic and environmental shocks interact with fixed or insecure incomes. Food and energy inflation erodes pensions and transfers when indexation is delayed. Older workers in informal employment may have neither retirement savings nor safe opportunities to continue work. Disasters destroy homes and assistive devices, interrupt medication, and create evacuation barriers; heat and air pollution amplify cardiovascular and respiratory risk. Yet Asia-Pacific evidence rarely reports disaster consumption, income, disability, or care outcomes separately for older adults. Mortality is well documented, while non-fatal capability loss and unpaid family-care burdens are not.

Figure 3 summarizes the life-cycle structure. Children have the largest duration multiplier, working-age adults the strongest immediate income and care channels, and older persons the highest mortality and health sensitivity. The groups are interdependent: an adult job loss changes a child's nutrition and an older relative's access to medicine, while bereavement or illness reallocates care and labour throughout the household.

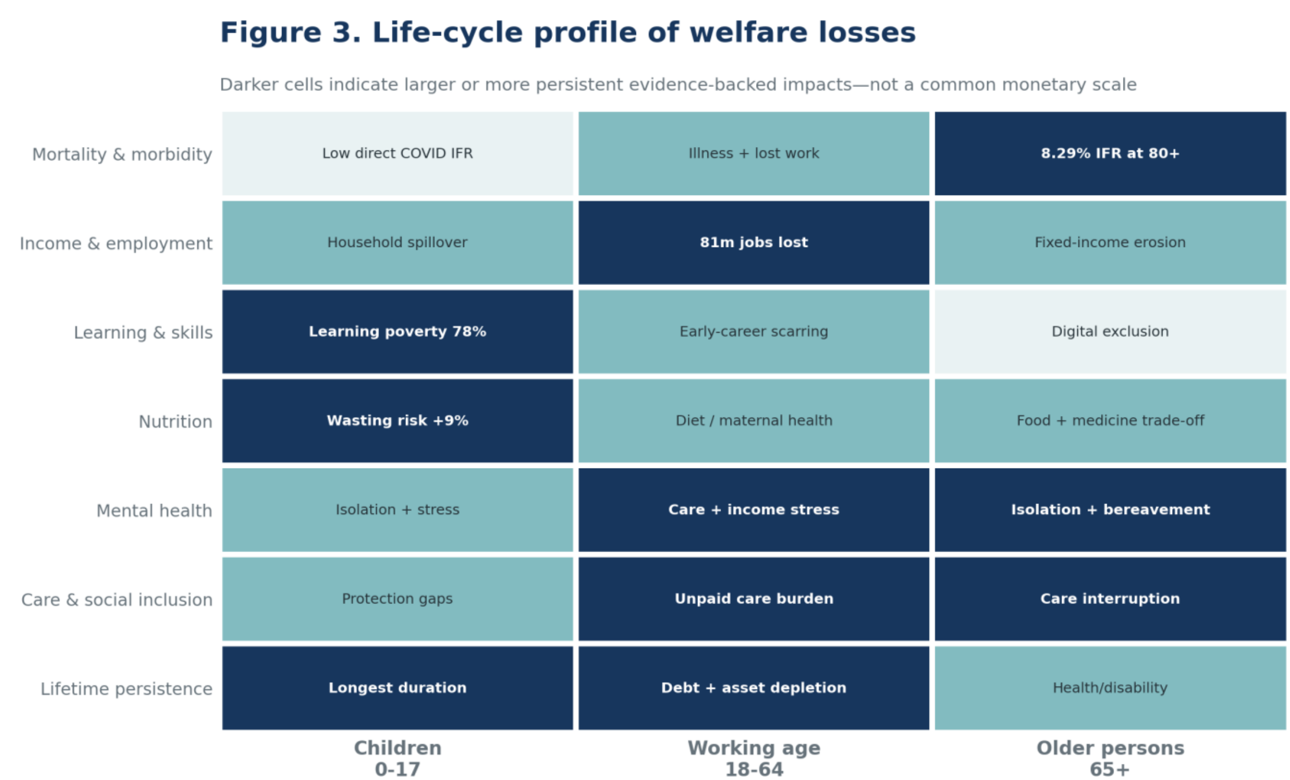

Figure 3. Life-cycle impacts among children, working-age adults, and older persons.

Source: authors' synthesis of the study-level evidence register.

Group	Primary welfare impacts	Quantitative estimates	Regions most affected
Children (0-17)	Learning, nutrition, service disruption, protection and psychosocial development	Learning poverty 60%→78% in South Asia; future earnings −up to 14.4%; wasting risk +9% after a 5% real food-price increase	South Asia; disaster-prone Southeast Asia and Pacific; pastoral Central/East Asia
Working-age adults (18-64)	Jobs, hours, earnings, informality, migration, debt, care and mental health	81 million Asia-Pacific jobs lost in 2020; women +8 pp work stoppage; Lao real wages about −33% in 2023	Region-wide; tourism-dependent Southeast Asia and Pacific; crisis economies
Older persons (65+)	Mortality, chronic disease, care access, isolation, fixed-income erosion and evacuation	Pre-vaccine IFR 8.29% at 80+; pandemic mortality concentrated at older ages; non-fatal Asia-specific estimates sparse	All subregions; highest gaps in low-capacity and conflict settings

Table 4. Welfare losses by demographic group.

7. Comparative magnitude, geography, and ranking

7.1 Which shocks were largest?

COVID-19 ranks first for short-run regional breadth and simultaneous damage across output, jobs, poverty, health, learning, nutrition, and mental health. The estimate of a 6.0-9.5% developing-Asia GDP loss in 2020 is larger than most regional estimates for recent inflation or trade shocks, while the poverty and job losses reached tens of millions (Asian Development Bank, 2021a, 2021b; International Labour Organization, 2020). No other event since 2015 affected almost every economy and welfare domain at once.

At country level, the largest proportional observed losses came from disasters and macroeconomic collapse. Cyclone Pam's damage and losses equalled 64% of Vanuatu's GDP, Cyclone Winston's 31% of Fiji's GDP, and the Tonga eruption's direct damage 18.5% of GDP. Sri Lankan poverty approximately doubled in one year, Afghanistan lost about one-fifth of GDP during the initial collapse, and Pakistan's floods threatened to add up to 9.1 million poor people. These cases cannot be ranked by a single number because asset damage, output, poverty, and mortality differ, but each exceeded the affected state's normal fiscal and household smoothing capacity.

Long-run climate change has the largest potential cumulative economic loss. ADB's high-end scenario reaches a 41% regional GDP loss by 2100 (Asian Development Bank, 2024). That magnitude carries wider uncertainty than observed pandemic or event accounting and should not be described as a forecast. The robust conclusion is ordinal: without mitigation and adaptation, repeated heat, sea-level, labour, agriculture, fisheries, and disaster losses compound rather than return automatically to baseline.

Figure 2 presents selected estimates in four panels instead of forcing them into one scale. GDP or purchasing-power losses are shown as percentages; people, jobs, and deaths as counts; human-capital outcomes as learning or earnings losses; and disaster burdens as percentages of national GDP. This preserves magnitude while making the unit differences visible.

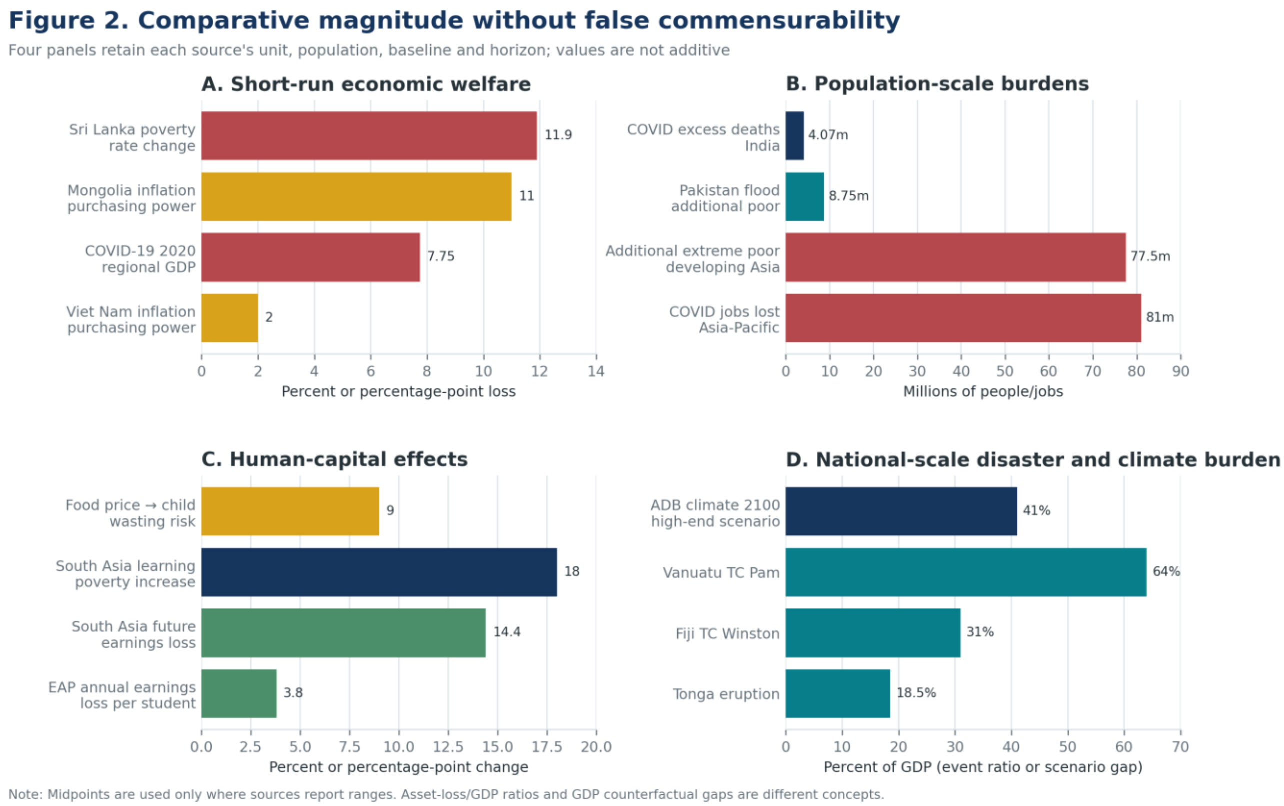

Figure 2. Comparative magnitude of selected welfare losses. Different panels retain different units.

Source: authors' synthesis of the study-level evidence register.

7.2 Which welfare dimensions were most affected?

Economic welfare has the largest volume of evidence: GDP, jobs, income, prices, poverty, and assets are routinely measured. Health losses are the most intrinsically severe, especially mortality, but comparable country estimates depend on registration and modelling. Human-capital losses are likely among the most persistent, because learning, stunting, and early-career scarring alter later productivity and mobility. Social welfare—agency, exclusion, violence, unpaid care, community cohesion, and loss of trust—is least consistently monetized or tracked.

The dimensions are not additive. A death already implies lost earnings and consumption; adding all three double counts. Lost schooling affects earnings through the same human-capital pathway. Asset loss reduces future consumption, while disaster reconstruction raises measured GDP. The appropriate aggregate would value changes in consumption, health, longevity, education, security, and distribution under one coherent social-welfare function. No reviewed regional study does this comprehensively.

7.3 Who was most vulnerable?

Poverty is the dominant cross-cutting vulnerability because it increases exposure and sensitivity while reducing coping. Poor households spend more on food, live in riskier locations, work in informal and climate-exposed jobs, own less insurable housing, and have less savings, connectivity, health care, and political voice. Children in such households show larger remote-learning gaps and larger nutrition responses to food prices. Women experienced higher pandemic work stoppage and carried unpaid care burdens. Migrants lost jobs and mobility. People with disabilities and displaced populations face greater evacuation and service barriers but are seldom separately quantified.

Vulnerability is also shock-specific. Older people and those with chronic disease faced the highest COVID-19, heat, and pollution mortality. Infants and pregnant women were most sensitive to nutrition and health-service shocks. Young workers were vulnerable to job separation and scarring. Farmers, fishers, herders, and outdoor workers faced climate and commodity exposure. Small island populations faced national-scale asset loss and limited diversification. A policy that is well targeted by income alone may therefore miss biological, occupational, geographic, and care vulnerabilities.

7.4 Geographic distribution

South Asia records the largest absolute human burdens in the reviewed evidence: pandemic excess mortality, prolonged school closure, learning poverty, child undernutrition, flood exposure, humid heat, air pollution, and Pakistan's flood losses. Population size partly explains the totals, but high baseline deprivation and exposure amplify them. Southeast Asia combines pandemic job and poverty losses with tourism dependence, food-price exposure, currency and political crises in selected economies, tropical cyclones, flood growth, and transboundary haze. East Asia has greater fiscal and infrastructure capacity on average, yet contains enormous flood exposure and China's pollution and trade risks. Central and West Asia face commodity dependence, food-price transmission, drought, conflict, displacement, and severe data gaps. The Pacific has smaller absolute populations but the largest disaster losses relative to GDP and exceptionally high fiscal and geographic concentration.

Figure 4 maps evidence coverage and selected high-severity outcomes by subregion. It is an evidence distribution, not a claim that a lightly studied country has low welfare loss. Data availability is endogenous to statistical capacity, language, conflict, donor presence, and the visibility of major events.

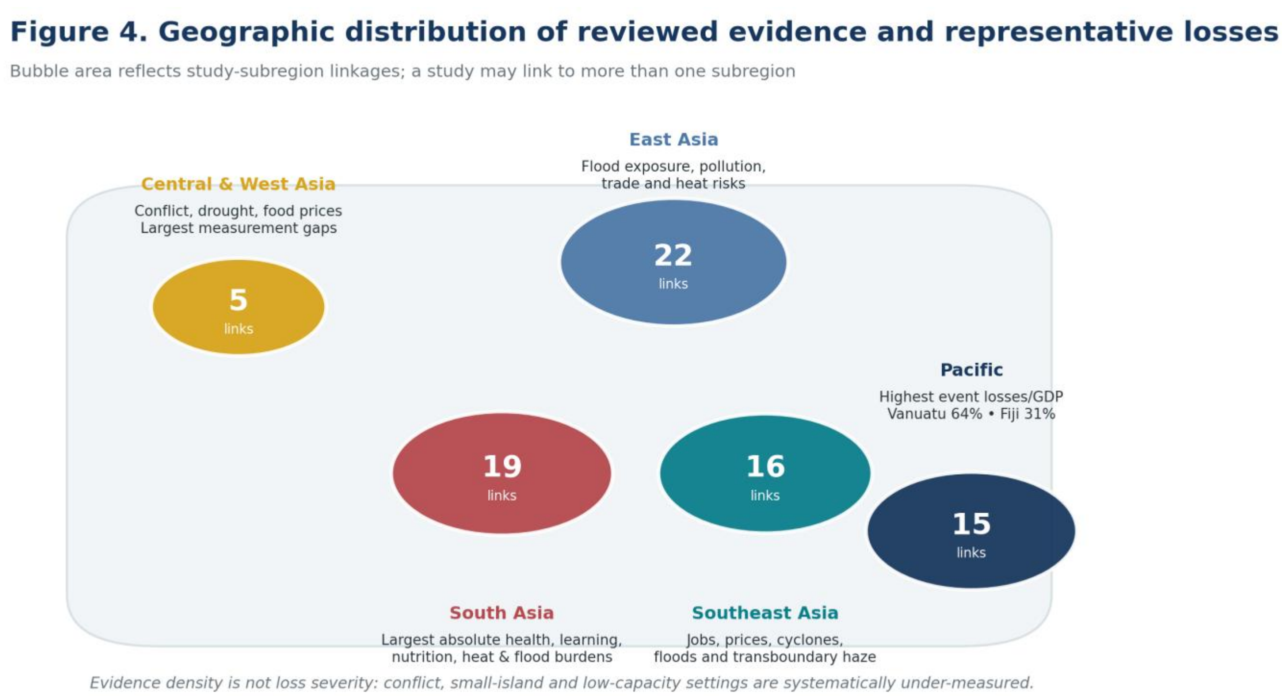

Figure 4. Geographic distribution of reviewed evidence and representative losses across Asia-Pacific subregions.

Source: authors' synthesis of the study-level evidence register.

Shock category	Short-term welfare losses	Long-term welfare losses	Evidence strength	Confidence assessment
COVID-19	Very high and region-wide: GDP, jobs, poverty, mortality, food security and services	High: excess mortality, learning, nutrition, debt and scarring	Large regional models, surveys, mortality and learning evidence	High for direction and ranking; medium for exact totals

Shock category	Short-term welfare losses	Long-term welfare losses	Evidence strength	Confidence assessment
Inflation and food/energy prices	Moderate regionally; very high for poor net consumers and crisis countries	Medium-high when nutrition, schooling or assets deteriorate	Good microsimulation; strong child-nutrition micro evidence	High for regressivity/mechanism; medium for national totals
Macro, debt, currency and trade shocks	Highly heterogeneous; severe in Sri Lanka, Afghanistan, Myanmar and Lao PDR	High if investment, services, female work and institutions remain impaired	Good crisis monitoring; attribution usually weak	Medium; low in conflict/data-collapse settings
Disasters and environmental degradation	Extreme locally; event effects up to 31%-64% of GDP in Pacific cases	High through assets, displacement, health, nutrition and repeated exposure	Strong PDNAs and causal health/productivity evidence; welfare aggregation sparse	High for event damage and key mechanisms
Long-run climate change	Already visible in heat mortality, productivity and disaster risk	Potentially the largest cumulative burden; ADB high-end GDP gap 41% by 2100	Multiple models agree on direction; magnitude varies with adaptation and damage functions	Medium for broad order; low for distant point estimates

Table 5. Comparative ranking of shocks. Rankings are ordinal and preserve differences in unit, horizon, and counterfactual.

8. Methodological challenges and evidence gaps

The first challenge is counterfactual construction. Pandemic GDP models compare outcomes with a pre-COVID path, while household surveys often compare a crisis month with recall or a pre-existing wave. Climate attribution compares observed weather with a modelled world without anthropogenic forcing. Post-disaster assessments compare replacement or repair costs with pre-event assets. Each counterfactual is legitimate for a specific question, but the resulting losses are not directly commensurate.

Second, welfare metrics differ in time and unit. A one-year GDP gap, a stock of destroyed assets, an undiscounted lifetime earnings loss, an excess death, and a person exposed to flood risk cannot be summed. GDP-equivalent valuation can improve comparison but requires value-of-statistical-life, discount-rate, inequality, and risk-aversion assumptions that are ethically and empirically contested. Purchasing-power-parity dollars aid cross-country comparison but do not reflect fiscal capacity or local replacement cost in the same way as market dollars.

Third, attribution is difficult under compound shocks. COVID-19 coincided with cyclones, drought, conflict, and commodity volatility. Sri Lanka's crisis combined debt, monetary and fiscal policy, shortages, global prices, and pandemic tourism losses. Pakistan's floods interacted with pre-existing inflation and fiscal stress. Climate change affects the probability or intensity of events rather than causing all damage independently; settlement patterns, land use, drainage, building quality, warning systems, and social protection govern consequences.

Fourth, household coping makes current consumption an incomplete measure. Borrowing, asset sales, skipped health care, lower diet quality, child labour, migration, and withdrawal from school can stabilize measured expenditure while reducing future welfare. Surveys rarely value unpaid care, informal insurance, lost leisure, psychosocial distress, or insecurity. Mortality statistics omit disability and bereavement; school-enrolment data omit learning; firm survival can conceal owner debt and worker downgrading.

Fifth, data systems underrepresent the most vulnerable. Phone surveys miss people without devices or stable connections. Conflict areas and small islands have sparse samples. Informal work and migration are poorly registered. Older persons, people with disabilities, indigenous communities, and people in institutions are rarely separately reported. Civil-registration gaps make mortality models depend on covariates precisely where shocks are most severe. High-income urban epidemiology is often transferred to rural or lower-income settings without local validation.

Sixth, publication and event-selection bias matter. Major disasters attract assessments; recurrent localized loss does not. Studies reporting statistically clear effects are more visible. Climate-economy projections can be unstable when data, damage functions, or code are corrected. This review excluded retracted evidence and treats long-horizon estimates as scenarios. A formal journal submission should add dual independent screening and extraction and refresh searches in Scopus, Web of Science, and EconLit through the ADB Library.

The highest-priority evidence gaps are therefore: panel data that follow households and children across shocks; consumption-equivalent disaster loss by income group; older-person income, disability, and care outcomes; gendered time use and violence; migration and remittance dynamics; effects on people with disabilities; small-island mental health and education; causal estimates of supply-chain and energy shocks; local value-of-statistical-life and quality-of-life weights; and integrated models that avoid double counting while representing inequality and capabilities.

9. Implications for resilience and policy design

A capability-based resilience strategy seeks to keep people above critical thresholds in food, health, learning, shelter, energy, mobility, care, and agency while preserving productive recovery. This requires ex ante systems rather than ad hoc relief. The evidence supports eight priorities.

First, social protection should be adaptive, interoperable, and capable of rapid vertical and horizontal expansion. Registries need links to civil identification, payments, labour, agriculture, disaster exposure, and local validation without excluding people who lack formal documentation or digital access. Trigger rules can use inflation, rainfall, flood, heat, employment, or health indicators. Transfers should be adequate to protect real consumption and indexed when prices move rapidly.

Second, food and nutrition protection should prioritize pregnant women, infants, young children, and poor households. Cash must be paired with functioning markets, treatment of wasting, school meals, micronutrients, maternal services, and climate-resilient food supply. The food-price evidence indicates that waiting for anthropometric deterioration is too late; price and diet indicators can trigger preventive support.

Third, learning recovery must be measured by competency rather than school reopening. Rapid assessment, structured pedagogy, tutoring, additional instructional time, teacher support, accessible digital content, and attendance outreach should target children with the largest losses. Earnings-loss estimates justify sustained remediation, but they should not be booked as inevitable if learning can be recovered.

Fourth, universal health coverage and public-health surveillance are economic resilience infrastructure. Continuity plans must protect routine immunization, maternal care, chronic-disease treatment, medicine supply, mental health, and elder care during emergencies. Excess-mortality monitoring requires stronger civil registration and cause-of-death systems. Heat-health action plans, air-quality alerts, occupational standards, clean household energy, and cooling access can prevent both mortality and productivity loss.

Fifth, macroeconomic stabilization should be distribution-aware. Targeted transfers generally protect poor households more efficiently than universal fuel or electricity subsidies, while lifeline tariffs can preserve essential energy access. Debt restructuring, exchange-rate adjustment, and fiscal consolidation should be sequenced with food, health, education, and employment protection. Excluding women from work is not only a rights loss but a macroeconomic loss.

Sixth, disaster-risk financing should be layered. Household savings and microinsurance cannot absorb national catastrophes. Contingency budgets, contingent credit, sovereign risk pools, catastrophe bonds where cost-effective, and predictable concessional finance should be matched to event frequency and severity. Financing must reach local governments and households quickly, while reconstruction standards reduce future exposure.

Seventh, adaptation should be embedded in investment appraisal, spatial planning, labour regulation, and public financial management. Flood protection, drainage, nature-based solutions, resilient housing, water security, cooling, early warning, and managed retreat have distributional consequences. Benefit-cost analysis should value avoided mortality, learning disruption, care burdens, and consumption volatility—not only avoided asset damage.

Eighth, resilience metrics should track the distribution and duration of loss. A core dashboard could report real consumption by quintile, food insecurity, job and hours loss, school competency, service continuity, excess mortality, mental health, asset depletion, displacement, and recovery time. Indicators should be disaggregated by sex, age, disability, location, migration status, and livelihood. The companion evidence workbook provides a prototype schema for keeping methods and source provenance visible.

10. Conclusion

The region's welfare losses from major shocks are multidimensional, unequal, and path dependent. COVID-19 produced the largest synchronized recent loss, combining a 2020 regional GDP gap of 6.0-9.5% with tens of millions of lost jobs and additional poor people, substantial excess mortality, and long-lived learning and nutrition damage. Inflation and macroeconomic crises were narrower geographically but highly regressive and, in Sri Lanka, Lao PDR, Afghanistan, Myanmar, and Mongolia, large enough to reverse years of progress. Environmental shocks generated the most extreme country-level asset and output ratios, particularly in Pacific island economies, while pollution and heat imposed large health and productivity burdens without a single disaster date. Climate change threatens to compound all of these pathways.

The most defensible comparison is therefore multidimensional rather than scalar. COVID-19 ranks highest in regional breadth; disasters and national collapses in local proportional severity; and climate change in potential cumulative duration. Children face the longest horizon of loss, older people the highest mortality risk, and working-age adults the central employment and coping burden. Poverty, informality, gender, disability, geography, and limited public protection convert exposure into welfare loss.

For policy, restoring GDP is insufficient. Recovery is complete only when capabilities, distribution, and resilience recover: children are learning, households can consume without destructive coping, workers regain secure livelihoods, older people receive care, preventable deaths fall, and the next shock does not reproduce the same losses. For measurement, the immediate priority is to link macro, household, health, education, climate, and administrative data around explicit counterfactuals and life-cycle outcomes. That is the empirical foundation required for a genuinely people-centred resilience agenda in Asia and the Pacific.

Figure notes and narrative descriptions

Figure 1. Conceptual pathways linking aggregate shocks to welfare losses. The figure traces epidemiological, economic, and environmental shocks through prices, labour markets, assets, public services, health exposure, ecosystems, and mobility. Household coping can buffer current consumption but transmit losses into debt, asset depletion, malnutrition, learning, mental health, and reduced resilience. Feedback arrows emphasize that today's coping affects exposure to tomorrow's shock.

Figure 2. Comparative magnitude of selected welfare losses. The figure uses four panels because units are not commensurate. Selected short-run GDP and purchasing-power estimates range from 2% to 11% for price shocks and 6.0-9.5% for the 2020 pandemic counterfactual; localized disaster damage and loss range from 18.5% to 64% of national GDP. Population-scale estimates reach 75-80 million additional people in extreme poverty and 81 million jobs lost during COVID-19. Human-capital panels show learning poverty rising 18 percentage points in South Asia and projected earnings losses of 3.8%-14.4% in selected regional studies. These quantities are displayed, not added.

Figure 3. Life-cycle impacts. A heat map contrasts the dominant mechanisms by age. Children receive the highest persistence rating for nutrition, learning, and lifetime earnings; working-age adults for jobs, earnings, migration, debt, and unpaid care; and older persons for mortality, chronic disease, social isolation, and care disruption. Inter-household arrows show that these burdens are jointly produced and shared.

Figure 4. Geographic distribution of evidence and loss. The figure reports the number of reviewed studies with explicit subregional estimates and annotates representative high-severity findings. South Asia has the densest evidence and the largest absolute mortality, learning, nutrition, heat, pollution, and flood burdens. Southeast Asia shows compound labour, price, cyclone, flood, and haze risks. Pacific economies exhibit the highest event losses relative to GDP. Central and West Asia has severe food, drought, conflict, and macroeconomic vulnerability but the largest measurement gaps.

Key quantitative estimates

The complete source-linked register is supplied in the companion workbook. The estimates that most strongly anchor the synthesis are: developing Asia's 6.0-9.5% GDP loss in 2020 relative to a no-COVID baseline; 75-80 million additional people in extreme poverty; 81 million Asia-Pacific jobs lost; 18.2 million global excess deaths in one peer-reviewed model, including 4.07 million in India; South Asian learning poverty rising from 60% to 78%; potential student earnings losses of up to 14.4% in South Asia; a 9% increase in child-wasting risk following a 5% real food-price rise; Sri Lankan poverty increasing from 13.1% to 25.0%; Lao real wages falling roughly 33% during the 2023 inflation shock; Pakistan's floods adding potentially 8.4-9.1 million poor people; single-event damage and losses of 31% of GDP in Fiji and 64% in Vanuatu; Indian manufacturing output falling around 2% per additional degree Celsius; 1.67 million air-pollution-attributable deaths in India in 2019; and a high-end scenario in which Asia-Pacific GDP is 41% lower by 2100. Each retains the source's baseline, horizon, and method.

Study	Key estimate	Estimate type	Confidence
Asian Development Bank (2021a)	6.0%-9.5% of regional GDP in 2020 and 3.6%-6.3% in 2021	Modelled counterfactual	Medium
Asian Development Bank (2021b)	75-80 million additional people in extreme poverty in 2020; roughly 170-180 million additional people below the broader poverty line	Poverty simulation	Medium
International Labour Organization (2020)	81 million jobs lost in 2020; working-hour losses pushed an additional 22-25 million workers into working poverty	Nowcast	Medium
COVID-19 Excess Mortality Collaborators (2022)	18.2 million global excess deaths in 2020-2021 versus 5.94 million reported COVID deaths; India 4.07 million, Indonesia 736,000, Pakistan 664,000	Excess mortality model	Medium
World Bank (2023a)	Learning poverty rose from about 60% to 78%; current students may lose up to 14.4% of future earnings; every 30 days of closure was associated with about 32 days of learning loss	Regional synthesis	Medium
Headey and Ruel (2023)	A 5% rise in real food prices over the previous three months increased wasting risk 9% and severe wasting 14%; the increase was 15% for asset-poor children and 6% for non-poor children	Large-sample microeconometric study	High
World Bank (2023b)	Poverty rose from 13.1% in 2021 to 25.0% in 2022 at the $3.65 2017 PPP line, adding about 2.5 million poor people; an estimated half a million industry and services jobs were lost, with annual average inflation of 46.4% in 2022	Country poverty nowcast	High
World Bank (2024a)	With 39% inflation in 2023 and nominal wages up 5.7%, real wages fell about 33%; in 2024 low-income real per-capita income fell 6.9%, and 36.2% of low-income households faced moderate or severe food insecurity	High-frequency household survey	Medium
Chodorow-Reich et al. (2020)	District-level cash shortages reduced employment and night-light output by at least 2 percentage points in 2016 Q4 and bank credit by about 2 points; effects dissipated over subsequent months	Quasi-experimental study	High
Government of Pakistan et al. (2022)	$14.9 billion in damage and $15.2 billion in economic losses; poverty projected to rise 3.7-4.0 percentage points, adding 8.4-9.1 million poor people; 1,730 deaths	Post-disaster needs assessment	High
Government of Vanuatu (2015)	About $450 million in damage and losses, equivalent to 64% of GDP; roughly 65,000 people displaced and livelihoods of at least 80% of the rural population compromised	Post-disaster needs assessment	High
Government of Fiji (2016)	44 deaths and about $1.38 billion in damage and losses, equivalent to 31% of GDP; 30,369 houses were damaged or destroyed and about 62% of the population was affected	Post-disaster needs assessment	High
Somanathan et al. (2021)	Annual plant output fell about 2% per 1 degree C increase; daily microdata show lower worker productivity and greater absenteeism, with climate control mitigating losses	Peer-reviewed microeconometric study	High
India State-Level Disease Burden Initiative Air Pollution Collaborators (2021)	1.67 million deaths in 2019 (17.8% of all deaths) were attributable to air pollution; output losses were $36.8 billion, or 1.36% of GDP	Burden-of-disease study	High
Asian Development Bank (2024)	Under a high-end emissions pathway, regional GDP could be 17% lower by 2070 and 41% lower by 2100; a Paris-aligned pathway limits the 2100 loss to around 11%; up to 300 million people face coastal-inundation risk	Integrated assessment scenario	Medium

Key quantitative estimates. Units, baselines and horizons follow the original sources; estimates are not additive.

Evidence confidence assessment

High-confidence findings are that COVID-19 caused very large employment, mortality, poverty, and learning losses; older people had sharply higher direct mortality risk; women, youth, less-educated, and informal workers experienced larger employment disruptions; food-price inflation increases child undernutrition risk; major disasters produce poverty increases well beyond direct asset damage; heat lowers labour productivity; and air pollution materially reduces life expectancy. These conclusions are supported by multiple data sources, strong designs, or official assessments.

Medium-confidence findings include the precise regional GDP counterfactual for COVID-19, the magnitude of lifetime earnings loss from school closures, nutrition deaths under service-disruption scenarios, purchasing-power losses from inflation microsimulations, welfare-equivalent annual disaster loss, and mid-century country climate losses. The mechanisms are well supported, but point estimates depend on model parameters and recovery assumptions.

Low-confidence findings concern precise household welfare totals in conflict-affected data environments, a unified ranking across incomparable welfare units, supply-chain-specific household losses, older-person non-fatal welfare losses, and end-century GDP point estimates. The report therefore uses ranges and ordinal conclusions rather than presenting a single regional welfare-loss total.

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Annotated bibliography: 20 influential studies

The studies below were selected for regional reach, empirical credibility, methodological influence, or importance to cross-shock comparison. Each note reports the study's quantitative contribution, design, interpretive value, and principal limitation.

1. Asian Development Bank (2021a) — ADB report

Coverage and finding. Developing Asia; All residents. The study reports 6.0%-9.5% of regional GDP in 2020 and 3.6%-6.3% in 2021.

Method and identification. Global macroeconomic scenario model with epidemiological and containment channels. The identifying basis is modelled counterfactual, not causal micro-identification.

Use and caution. It is influential here because it supplies a modelled counterfactual for gdp relative to a no-pandemic baseline. Principal limitation: Range depends on pandemic and policy scenarios; GDP is not a complete welfare metric. Confidence: Medium. Source

2. Asian Development Bank (2021b) — Key Indicators for Asia and the Pacific 2021

Coverage and finding. Developing Asia; People near international poverty lines. The study reports 75-80 million additional people in extreme poverty in 2020; roughly 170-180 million additional people below the broader poverty line.

Method and identification. Growth-to-poverty simulation against a no-COVID baseline. The identifying basis is model-based nowcast.

Use and caution. It is influential here because it supplies a poverty simulation for extreme and moderate poverty. Principal limitation: Sensitive to poverty line, survey vintage, distributional pass-through, and baseline growth. Confidence: Medium. Source

3. International Labour Organization (2020) — Asia-Pacific Employment and Social Outlook 2020

Coverage and finding. Asia and the Pacific; Working-age population. The study reports 81 million jobs lost in 2020; working-hour losses pushed an additional 22-25 million workers into working poverty.

Method and identification. ILO nowcasting model combining labour-force, national accounts, and high-frequency data. The identifying basis is modelled deviation from pre-crisis expectations.

Use and caution. It is influential here because it supplies a nowcast for employment and working poverty. Principal limitation: Informal work and hours are imputed where real-time data are incomplete. Confidence: Medium. Source

4. Egger et al. (2021) — Science Advances

Coverage and finding. Nine LMICs, including Bangladesh, Nepal, and the Philippines; More than 30,000 households across 16 samples. The study reports Income fell in 8%-87% of samples (median about 70%); employment fell 5%-49% (median 30%); 9%-87% missed or reduced meals (median 45%).

Method and identification. Harmonized rapid phone surveys linked to pre-pandemic samples. The identifying basis is before-after descriptive comparisons across multiple cohorts.

Use and caution. It is influential here because it supplies a multi-country panel survey for income, employment, food security, and market access. Principal limitation: No unexposed control; samples and questions are heterogeneous; phone attrition may be selective. Confidence: Medium. Source

5. World Bank (2021b) — Long COVID: East Asia and Pacific Economic Update

Coverage and finding. Developing East Asia and Pacific; Students. The study reports Average annual earnings loss of about $524 per student in 2017 PPP, or 3.8%; estimated loss reaches 9.3% in ASEAN-5 and 7.9% in small East Asian economies.

Method and identification. Learning-adjusted schooling losses mapped to returns to education and lifetime earnings. The identifying basis is simulation using closure, learning, and earnings parameters.

Use and caution. It is influential here because it supplies a human-capital simulation for lifetime earnings. Principal limitation: Long-run projection; recovery, remediation, discount rates, and labour-market returns are uncertain. Confidence: Medium. Source

6. World Bank (2023a) — Collapse and Recovery: How COVID-19 Eroded Human Capital and What to Do About It

Coverage and finding. South Asia; Children and young people. The study reports Learning poverty rose from about 60% to 78%; current students may lose up to 14.4% of future earnings; every 30 days of closure was associated with about 32 days of learning loss.

Method and identification. Regional synthesis of assessment data, closure duration, and human-capital simulations. The identifying basis is observed assessments plus modelled lifetime effects.

Use and caution. It is influential here because it supplies a regional synthesis for learning poverty and future earnings. Principal limitation: Some countries lack comparable post-pandemic tests; earnings effects are projections. Confidence: Medium. Source

7. Osendarp et al. (2021) — Nature Food

Coverage and finding. 118 low- and middle-income countries, with heavy burden in South Asia; Children under five and women. The study reports Moderate 2020-2022 scenario: 9.3 million additional wasted children, 2.6 million additional stunted children, 168,000 additional under-five deaths, and $29.7 billion in future productivity losses.

Method and identification. Linked MIRAGRODEP, nutrition-risk, LiST, and Optima Nutrition models. The identifying basis is scenario modelling.

Use and caution. It is influential here because it supplies a nutrition scenario model for wasting, stunting, maternal nutrition, deaths, and productivity. Principal limitation: Not Asia-only; results depend on assumed income and service shocks and may overlap with other pandemic estimates. Confidence: Medium. Source

8. COVID-19 Mental Disorders Collaborators (2021) — The Lancet

Coverage and finding. 204 countries and territories, including Asia-Pacific; All ages; sex- and age-disaggregated. The study reports 53.2 million additional major-depressive-disorder cases (+27.6%) and 76.2 million additional anxiety cases (+25.6%) in 2020; females and younger people bore larger increases.

Method and identification. Systematic review and Bayesian meta-regression linked to mobility and infection indicators. The identifying basis is model-based association using pre-post prevalence studies.

Use and caution. It is influential here because it supplies a global burden model for major depressive and anxiety disorders and dalys. Principal limitation: Sparse direct surveys in many Asian countries; no clean separation of infection, restrictions, and recession channels. Confidence: Medium. Source

9. COVID-19 Excess Mortality Collaborators (2022) — The Lancet

Coverage and finding. 191 countries and territories; All ages. The study reports 18.2 million global excess deaths in 2020-2021 versus 5.94 million reported COVID deaths; India 4.07 million, Indonesia 736,000, Pakistan 664,000.

Method and identification. All-cause mortality comparison plus ensemble models for data-poor locations. The identifying basis is excess-mortality counterfactual.

Use and caution. It is influential here because it supplies a excess mortality model for excess deaths. Principal limitation: Only 74 countries had full mortality series; model uncertainty is substantial in data-scarce settings. Confidence: Medium. Source

10. Kugler et al. (2023) — World Development

Coverage and finding. 40 developing countries, including East and South Asia; Working-age adults. The study reports Women were 8 percentage points more likely than men to stop work; youth 4 points more likely than older adults; low-education and urban workers each about 4 and 3 points more likely.

Method and identification. Harmonized high-frequency phone surveys with multivariate decomposition. The identifying basis is cross-sectional conditional gaps during the crisis.

Use and caution. It is influential here because it supplies a harmonized household surveys for work stoppage by sex, age, education, and location. Principal limitation: Not causal treatment effects; phone samples and timing differ; country averages conceal informal-work intensity. Confidence: Medium. Source

11. Headey and Ruel (2023) — Nature Communications

Coverage and finding. 44 low- and middle-income countries, including South and Southeast Asia; 1.27 million children under five. The study reports A 5% rise in real food prices over the previous three months increased wasting risk 9% and severe wasting 14%; the increase was 15% for asset-poor children and 6% for non-poor children.

Method and identification. DHS microdata merged with local food-price series; child, season, and location controls. The identifying basis is within-country observational panel of repeated cross-sections.

Use and caution. It is influential here because it supplies a large-sample microeconometric study for wasting and stunting. Principal limitation: Price endogeneity and incomplete local market coverage remain; estimates are relative risks, not percentage-point changes. Confidence: High. Source

12. World Bank (2023b) — Sri Lanka Development Update: Time to Reset

Coverage and finding. Sri Lanka; Households and workers. The study reports Poverty rose from 13.1% in 2021 to 25.0% in 2022 at the $3.65 2017 PPP line, adding about 2.5 million poor people; an estimated half a million industry and services jobs were lost, with annual average inflation of 46.4% in 2022.

Method and identification. Household survey nowcasting, labour indicators, and macro-poverty simulation. The identifying basis is observed crisis period with modelled poverty update.

Use and caution. It is influential here because it supplies a country poverty nowcast for poverty, jobs, living costs, and nutrition. Principal limitation: The pandemic, policy errors, global prices, and debt crisis are jointly determined; no single-shock causal decomposition. Confidence: High. Source

13. Chodorow-Reich et al. (2020) — Quarterly Journal of Economics

Coverage and finding. India; Workers, firms, and districts. The study reports District-level cash shortages reduced employment and night-light output by at least 2 percentage points in 2016 Q4 and bank credit by about 2 points; effects dissipated over subsequent months.

Method and identification. Cross-district instrumental variables using pre-existing currency denomination exposure. The identifying basis is quasi-experimental iv.

Use and caution. It is influential here because it supplies a quasi-experimental study for employment, night-light output, and credit. Principal limitation: Captures short-run local real effects; informal transactions and broader national effects are imperfectly measured. Confidence: High. Source

14. Hallegatte et al. (2017) — World Bank, Unbreakable

Coverage and finding. 117 countries, including Asia-Pacific; Households differentiated by income and resilience. The study reports $520 billion in annual global consumption-equivalent wellbeing losses and 26 million people pushed into extreme poverty each year; resilience policies could cut wellbeing losses about 20%.

Method and identification. Asset-loss risk model linked to household consumption, vulnerability, and coping capacity. The identifying basis is ex ante welfare-risk model.

Use and caution. It is influential here because it supplies a welfare-risk model for consumption-equivalent wellbeing and poverty. Principal limitation: Global headline is not an Asia subtotal; depends on resilience and consumption-smoothing assumptions. Confidence: Medium. Source

15. Asian Development Bank (2024) — Asia-Pacific Climate Report 2024

Coverage and finding. Developing Asia and the Pacific; Regional economies and exposed populations. The study reports Under a high-end emissions pathway, regional GDP could be 17% lower by 2070 and 41% lower by 2100; a Paris-aligned pathway limits the 2100 loss to around 11%; up to 300 million people face coastal-inundation risk.

Method and identification. Integrated sectoral damage modelling with macroeconomic aggregation. The identifying basis is long-run scenario comparison.

Use and caution. It is influential here because it supplies a integrated assessment scenario for gdp, labour productivity, coastal assets, and poverty risk. Principal limitation: Deep uncertainty in climate, adaptation, tipping points, and damage functions; not a forecast. The recorded URL is the report's landing page, not the report PDF; the US$300 billion adaptation-finance figure could not be located in reachable text. Source URL must be replaced before that figure is quoted (CONSTITUTION.md §2.7).. Confidence: Medium. Source

16. Diffenbaugh and Burke (2019) — Proceedings of the National Academy of Sciences

Coverage and finding. Global, with country results for Asia; National populations. The study reports Warming increased population-weighted between-country inequality by about 25%; per-capita GDP was reduced 17%-31% at the poorest four deciles of the population-weighted country distribution, and the 18 countries with the lowest cumulative per-capita emissions had a median impact of about -27% relative to a no-warming counterfactual.

Method and identification. Detection-and-attribution climate simulations combined with nonlinear temperature-growth response. The identifying basis is historical climate-attribution counterfactual.

Use and caution. It is influential here because it supplies a attribution model for gdp per capita and between-country inequality. Principal limitation: Depends on contested growth response and cumulative counterfactual; estimates are not household consumption losses. The previously recorded country figures (31% India, 27% Indonesia, 12% Bangladesh) are not supported by this paper: Indonesia and Bangladesh appear nowhere in its text, and the 31% and 27% figures are a decile range and a low-emitter median respectively, not country estimates (CONSTITUTION.md §2.7). Confidence: Medium. Source

17. Carleton et al. (2022) — Quarterly Journal of Economics

Coverage and finding. Global, including India and Asia-Pacific; All ages. The study reports Under high emissions, net mortality costs including adaptation were valued at about 3.2% of global GDP in 2100; the mortality partial social cost of carbon was $36.6 per tCO2.

Method and identification. Subnational mortality-temperature panel, age-specific response, adaptation, and valuation model. The identifying basis is reduced-form panel plus forward projection.

Use and caution. It is influential here because it supplies a peer-reviewed econometric projection for mortality and monetized mortality risk. Principal limitation: Covers mortality only; value-of-statistical-life and adaptation assumptions are consequential. Confidence: High. Source

18. Somanathan et al. (2021) — Journal of Political Economy

Coverage and finding. India; Manufacturing workers and plants. The study reports Annual plant output fell about 2% per 1 degree C increase; daily microdata show lower worker productivity and greater absenteeism, with climate control mitigating losses.

Method and identification. Worker-level production records and national plant panels with weather variation. The identifying basis is reduced-form panel using short-run temperature fluctuations.

Use and caution. It is influential here because it supplies a peer-reviewed microeconometric study for worker productivity, absenteeism, and plant output. Principal limitation: Selected firms for worker microdata; results may not generalize to all sectors or adaptation margins. Confidence: High. Source

19. Rentschler, Salhab, and Jafino (2022) — Nature Communications

Coverage and finding. 188 countries; Population exposed to 1-in-100-year floods. The study reports 1.81 billion people were directly exposed globally; 1.24 billion lived in South and East Asia, including 395 million in China and 390 million in India; 170 million exposed people lived in extreme poverty.

Method and identification. High-resolution flood maps overlaid with population and subnational poverty surfaces. The identifying basis is exposure mapping.

Use and caution. It is influential here because it supplies a geospatial exposure assessment for flood exposure by poverty status. Principal limitation: Exposure is not expected loss; defenses, depth, duration, coping, and actual event probability vary. Confidence: High. Source

20. Government of Pakistan et al. (2022) — Pakistan Floods 2022 Post-Disaster Needs Assessment

Coverage and finding. Pakistan; 33 million affected people. The study reports $14.9 billion in damage and $15.2 billion in economic losses; poverty projected to rise 3.7-4.0 percentage points, adding 8.4-9.1 million poor people; 1,730 deaths.

Method and identification. Sectoral PDNA, household exposure, and macro-poverty simulation. The identifying basis is post-disaster accounting and modelled human impact.

Use and caution. It is influential here because it supplies a post-disaster needs assessment for damage, economic losses, poverty, and multidimensional poverty. Principal limitation: Rapid assessment; indirect and long-run health, learning, and displacement losses are incomplete. Confidence: High. Source

Review protocol and reproducibility note

The companion workbook contains one row per quantitative source, the required extraction fields, confidence coding, source URLs, and the data used for Figures 2-4. The manuscript and figures were generated from the same structured register so that numeric updates can be traced.

This is a structured rapid evidence review with systematic-scoping elements, not a registered systematic review. Before external journal submission, the search should be refreshed in Scopus, Web of Science, EconLit, PubMed, and the ADB Library; titles, abstracts, and full texts should be dual-screened; and a second reviewer should independently verify all extracted estimates.

Version: 2026-08-04. Evidence cutoff: 2026-07-31. Main narrative word count: approximately 9,655 words, excluding references, tables, and the annotated bibliography.